FOLLOW MY VOICE: ASSISTED GAZE-CONTINGENT SEARCH WITH A SELF-SIMILAR AGENT IN VIRTUAL REALITY

by

PEDRO POVEDA QUEVEDO
B.Sc. in Computer Science

A thesis submitted in partial fulfilment of the requirements
for the degree of Master of Science
in the Department of Computer Science
in the College of Engineering and Computer Science
at the University of Central Florida
Orlando, Florida

Fall Term
2026

Major Professor: Dr. Roshan Venkatakrishnan

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ABSTRACT

Directing users' attention to objects of interest has long been a concern in virtual reality (VR), particularly when relevant objects lie outside the user's current field of view [17, 30]. Beyond visual cues, researchers have investigated audio alerts for visual targets in VR [43], spatialized sound for out-of-view targets in augmented reality (AR) [5], and spoken hints for AR object search [54]. Spoken assistance can connect spatial instructions with objects in the surrounding environment, guiding users toward a region and then the intended object. Such guidance also has accessibility applications: verbal directions have helped blind users locate points of interest on physical objects [11]. The experience of receiving assistance can also be influenced by the tone and voice identity of spoken instructions. Research on virtual collaboration has found that self-similar voices can increase likability and believability but can also elicit greater eeriness [19]. As interactive agents take on a more active role in guiding users through virtual environments, the experience of receiving their assistance becomes an important design consideration. These findings motivate examining how self-similarity affects users when the voice serves as a spatial guide. This thesis compares gaze-contingent guidance spoken in a synthetic cloned voice resembling the participant's own (self-similar) with guidance spoken in a neutral voice selected by the participant from a supplied catalog.

Follow My Voice investigates this comparison in a within-participant virtual-reality user study. Participants complete one practice trial in their first assigned voice and two counterbalanced blocks of 40 measured trials. Seated participants search for colored geometric objects arranged on eight vertical planes around them in a virtual environment, including locations initially outside their field of view. Initially, participants are not informed of which object they should find. Instead, an agent that uses the participant's gaze data as context guides them toward it with spatial hints such as "turn to the left" or "look up." Participants select the target by pointing the right controller at the object

and pressing the trigger. Both trial conditions use the same hint guidance to help participants find the object.

The study compares performance in terms of active search-completion time, alongside perceived workload, motivation, and agent experience. Condition-specific questionnaires are administered after both blocks.

ACKNOWLEDGMENTS

I would like to thank Dr. Roshan Venkatakrishnan for his guidance and support throughout the development of this thesis. I also extend special thanks to Rohith Venkatakrishnan for his support as well.

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LIST OF ABBREVIATIONS

AI Artificial intelligence

API Application programming interface

AR Augmented reality

ARM64 64-bit Arm architecture

CSV Comma-separated values

EEG Electroencephalography

EQ1 Engineering question 1

ETD Electronic thesis or dissertation

ID Identifier

IMI Intrinsic Motivation Inventory

JSON JavaScript Object Notation

NASA-TLX National Aeronautics and Space Administration Task Load Index

PDF Portable Document Format

SVG Scalable Vector Graphics

TLX Task Load Index (see NASA-TLX)

UCF University of Central Florida

VR Virtual reality

XR Extended reality

CHAPTER 1: INTRODUCTION

1.1 Motivation and Background

Directing a user's attention to objects of interest is a persistent challenge in virtual reality (VR). An immersive environment can surround the user with relevant content, such as targets, instructions, or points of interest, yet only the portion within the current field of view is visible at any moment. Content outside that view must first be brought to the user's attention before it can be inspected or acted on. Prior work has addressed this through subtle gaze guidance that steers attention across an immersive scene [17], through motion cues that direct attention while preserving immersion [30], and through audio alerts that draw attention to objects of interest [43]. These approaches share a common design problem: how can a system help users locate relevant content without pulling them out of the surrounding environment?

1.1.1 Guiding Users Toward Objects of Interest

The present study explores how the voice used to deliver instructional guidance influences task performance, perceived workload, and enjoyment. It compares self-similar and neutral voices while keeping the content and mechanism of guidance consistent across conditions.

Binetti et al. examine augmented-reality (AR) search assistance motivated by maintenance, repair, instruction, and training, settings in which a person must locate a target in the surrounding scene [5]. Zhang et al. examine the concrete example of finding a specified book among other books [54]. Such tasks let researchers study how users connect a cue with its intended object without reproducing an entire workplace activity.

Chiossi et al.'s *Searching Across Realities* also studies object search, providing a methodological rationale for using it as a controlled task with behavioral and gaze measures [9]. Their study examines how the presentation of physical and virtual content affects search efficiency and attention allocation. For the present comparison, this work motivates specifying the visual environment carefully so that the guiding voice can be varied within a consistent search scene.

When an object begins outside the field of view, assistance must first support attention orientation, directing the user toward a relevant region of the scene [43]. Once that region becomes visible, the user still needs to distinguish the intended object from nearby distractors.

1.1.2 From Auditory Cues to Spoken Assistance

Research on auditory guidance supports using sound to help users locate objects. In head-mounted AR, Binetti et al. combined visual arrows with spatialized alarm sounds to guide users toward out-of-view targets [5].

A VR exhibition study compared a rearview mirror, a proximity beep, and a controller vibration against a no-cue control while participants searched for seven target paintings [43]. Audio and haptic alerts reduced how often participants consulted their target list, yet neither shortened completion time relative to the control. These alerts carried minimal information, signaling only that a target was nearby. Spoken guidance can instead specify where to look, which motivates examining speech as a richer channel for orienting attention.

Previous studies have examined using spoken hints to help identify an object. Zhang et al.'s AR book-search study used an audio instruction identifying an approximate shelf [54].

Other studies have used the user's gaze data to generate tonal audio that guides them toward a target object. Losing et al. varied a tone with gaze-to-target distance, and Kwok et al. combined initial

spatial-audio orientation with finer guidance [33, 29]. Both studies used desktop displays rather than immersive headsets, and their nonverbal cues differ from conversational speech. Their relevance here is the principle of linking the user's current gaze to the information the system supplies next.

Follow My Voice applies this principle through short spoken directions that first orient the participant toward a target region and then refine the search within it. Because the agent knows the target while the participant initially does not know which object to select, the participant depends on the agent's gaze-based instructions to find it. The participant therefore hears the guiding voice repeatedly during search, which raises a further question: does hearing that guidance in a voice resembling one's own change the experience or effectiveness of receiving help?

1.1.3 Self-Similar Voice and the Doppelgänger Connection

The voice delivering an instruction can influence how users experience an interaction as well as how they engage with a task. Studies of self-similar voices have reported differences in game performance, immersion, and perceptions of a virtual collaborator [25, 19]. In Follow My Voice, self-similar synthesis creates a cloned voice resembling the participant, allowing the study to examine whether that resemblance changes the experience or effectiveness of receiving spatial instructions.

Kao et al. provide evidence that voice resemblance can affect both behavior and experience [25]. In an online programming game with 657 analyzed participants, players assigned a self-similar avatar voice completed more puzzles and played for longer than those assigned a dissimilar voice. They also reported greater similarity identification, competence, relatedness, and immersion, although direct differences in the measured intrinsic-motivation subscales were not significant. Performance in that study meant the number of puzzles completed during voluntary play, rather than faster completion of a fixed task.

Guo et al.'s *Collaborating with my Doppelgänger* brings this question into immersive VR [19]. Their 25-participant study crossed self-similar appearance and voice while participants solved a jigsaw puzzle with a virtual character. Self-similar voices increased the character's likability and believability, but also elicited greater eeriness. This combination of favorable impressions and discomfort is particularly relevant to a guide that speaks repeatedly throughout a task.

Evidence from voice listening also connects resemblance with social judgments. Jaggy et al. found that recordings of other speakers whose voices were more similar to a listener's own tended to receive higher likability and trustworthiness ratings [23]. Because this experiment used recorded human voices rather than cloned agents delivering instructions, it supports a possible link between vocal resemblance and agent impressions, rather than demonstrating improved task assistance.

Voice choice offers a related, but distinct, form of personalization. In *Audio Matters Too*, Kao et al. found that choosing an avatar's voice improved identification and perceived autonomy when participants could also choose its visual appearance; those benefits were not observed without visual choice [24]. Follow My Voice compares a self-similar cloned voice with a neutral voice selected by the participant from the available options. Here, *neutral voice* denotes an existing voice not cloned from the participant; it does not imply emotionally flat speech or an absence of preference. The comparison asks whether self-similar synthesis offers value beyond a voice the user has already chosen, while recognizing that the two voices can differ in naturalness and other acoustic properties as well as resemblance.

The present study extends this literature from an avatar or embodied collaborator to a voice that guides search for an initially unnamed object. Participants use its spatial instructions to determine where to look and what to select, while the agent has no visible body. Perceived voice-similarity ratings assess the intended resemblance; search completion time, workload, motivation, and agent experience address the consequences of receiving guidance in each voice.

1.1.4 A Controlled Setting for Self-Similar Assistance

Follow My Voice examines this comparison through seated object search in virtual reality. Colored geometric objects are distributed across eight vertical planes around the participant. Some targets initially lie outside the participant's field of view. The application knows which object is the target, while the participant is not given its color or shape at the start. Spoken directions first guide orientation toward the relevant wall and then refine the search within it. The participant aims a right-controller ray and presses the trigger to select the inferred target; gaze itself does not select an object.

Simple geometric shapes on a common arrangement of surrounding planes make starting geometry and trial events easier to specify consistently across the two voice conditions. Object search gives the speech a concrete purpose with observable outcomes, and because the target is initially unnamed, guidance remains consequential until the participant selects it.

The central comparison varies the assigned voice while keeping the assistance mechanism common to both conditions. Gaze contingency and constrained agent decisions use the current search state to select instructions.

Spoken guidance also offers a relevant direction for low-vision assistance because spatial instructions can convey information that would otherwise rely on visible markers. Coughlan et al.'s work with blind users demonstrates the feasibility of verbal corrections for locating points of interest on physical objects [11]. The present method offers a controlled way to study such instruction, but its visual targets and gaze-dependent feedback require separate accessibility testing. An extension for people with low vision would need to consider residual vision, target contrast and size, tracking reliability, and alternatives to gaze-dependent readiness and controller aiming.

1.2 Problem Statement

Previous research explores how visual and auditory guidance can direct attention toward objects of interest, including targets outside the user's current field of view [17, 30, 5, 54]. A related body of work examines how self-similar voices influence the experience of interacting with a virtual agent, including favorable impressions and eeriness [19, 25]. Together, these areas motivate examining self-similar spoken assistance in a task where instructions help users discover which object they are seeking.

This study explores the impact of self-similar spoken assistance when guiding search for an initially unnamed object in virtual reality. It evaluates how receiving spatial instructions in a self-similar synthetic voice, compared with a participant-selected neutral voice, affects task performance and the experience of performing the search. The comparison considers active search-completion time, perceived workload, motivation, and agent experience while keeping the task and guidance mechanism consistent across conditions. Because the voices can also differ in naturalness and other acoustic properties, resemblance is not the only difference between conditions.

Answering this question requires appropriate guidance in both conditions and a record of what each participant heard. Gaze contingency, the shared decision policy, and playback logging address this requirement. Interpreting the interaction also requires independent gaze-quality checks and separate records of looking, pointing, and selecting [15, 40]. Chapter 3, including Section 3.4, describes these controls.

1.3 Research Objectives and Questions

The primary objective is to evaluate self-similar voice as a form of task assistance, using a neutral voice as the comparison within a shared gaze-contingent system. The primary research question is: does active search-completion time differ between self-similar and neutral voices after accounting for absolute starting angle and repeated observations within participants? The contrast is two-sided because no performance advantage for self-similarity is assumed.

Secondary questions concern perceived workload and activity experience. Will the two voices differ on the six workload dimensions and their complete unweighted mean? Will they differ on interest/enjoyment, effort/importance, pressure/tension, or value/usefulness? Because these outcomes concern different constructs, the analysis reports them separately rather than pooling them into a broad satisfaction score.

Exploratory questions concern agent perception, reliance, and the relationship between task difficulty and voice experience. Participants may describe a voice as more responsive or humanlike, may rely on guidance more often, or may experience a cloned voice as familiar or eerie.

A supporting objective is to make the assistance process inspectable so that technical faults can be separated from effects of self-similar voice. Can recorded state and speech history explain why a cue was eligible, why it played or was suppressed, and how long it waited? Can the evaluation distinguish an unhelpful instruction from an acceptable alternative wording?

1.4 Scope of the Comparison

The independent variable is voice condition: neutral voice versus self-similar synthetic voice. Both conditions use gaze contingency, the same semantic phrase library, the same local geometry rules,

and the same controller-selection mechanics. Because these properties are shared, the study cannot estimate the separate contributions of gaze contingency, short-term memory, the cloud decision agent, or synthetic speech in general.

The task is deliberately narrower than unrestricted navigation. Participants remain seated and rotate their head and chair to search surrounding planes. Guidance such as "turn left" refers to orientation within this seated setup rather than walking toward a target. Objects appear in a virtual room, and the target is initially unnamed. Conclusions are limited to guided search under these conditions.

The single practice trial uses the first assigned voice, so whichever voice appears first receives additional exposure. Counterbalancing distributes this asymmetry across voices but does not remove learning or carryover; the analysis therefore models period and assigned order, and the practice assignment is reported.

Figure 1.1 separates the voice manipulation from the shared task.

1.5 Intended Contributions

The thesis develops an empirical comparison of self-similar voice with a preferred alternative in virtual-reality task assistance. Its conceptual contribution is to distinguish resemblance from preference and to examine performance alongside the favorable or uncomfortable experience of hearing one's own synthetic voice.

The methodological contribution is a within-participant design for this comparison: two counterbalanced voice blocks, a shared hidden-target task, explicit trigger selection, identical nominal starting-angle quotas across blocks, retrospective voice-referenced questionnaires, and a planned

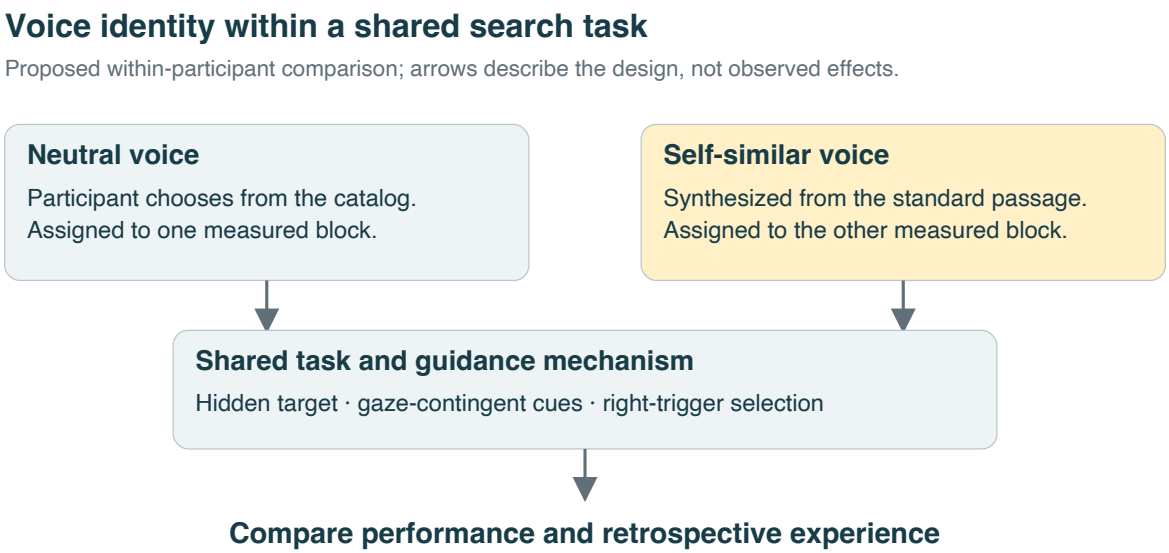


Figure 1.1: Participants complete the same search task twice, once with a neutral voice and once with a self-similar voice. Only the voice changes. The gaze-contingent hint logic, phrase library, starting-angle allocation, and selection method are the same in both conditions, so outcome differences reflect the voice configuration rather than the guidance method.

repeated-measures analysis. Manipulation checks and acoustic-quality reporting characterize what each voice condition delivered before outcomes are interpreted.

The guidance and recording platform serves as the apparatus for this comparison. Local spatial eligibility, constrained agent decisions, prepared speech, and synchronized review allow inspection of what each participant heard and whether delivery problems affected the comparison. Neither a provider benchmark nor a new gaze-guidance algorithm is claimed as a primary contribution.

The empirical contribution is pending; the thesis does not yet report a recruited sample, condition means, significance tests, or participant quotations.

CHAPTER 2: LITERATURE REVIEW

AR and VR are being investigated across an expanding range of spatial tasks, from finding objects in outdoor environments to receiving instructions tied to a surrounding scene. This review first situates the study among these applications, then examines voice identity and self-similarity, attention guidance, and the gaze-based interaction that supports the present comparison.

2.1 AR and VR Applications for Spatial Assistance

Kumaran et al. compare compass, radar, and in-world arrow aids during outdoor AR search for virtual objects among physical and virtual scene content [28]. At a different scale, Yang et al.'s AR navigation preprint combines building information with an embodied guide and directional arrows to assist navigation queries [50].

Other work addresses spatial instructions within a task rather than movement toward a distant destination. Binetti et al. investigate locating out-of-view objects in head-mounted AR, while Zhang et al. study visual and auditory hints during AR search [5, 54]. VR offers a complementary setting in which the arrangement of targets, distractors, and initial viewing directions can be controlled and recorded. Chiossi et al.'s comparison of search across physical and virtual content further connects object search with measurements of efficiency and attention [9].

Follow My Voice concentrates on the voice delivering spatial instructions: it compares self-similar and neutral speech within a common VR search application, rather than comparing AR with VR or replacing a navigation interface.

2.2 Voice as a Social and Personalized Interface

Research on machine-generated speech has considered personality, source attribution, and social presence [38, 37, 31]. These perspectives motivate treating voice as more than a neutral carrier of words.

In this project, personalization takes two forms. In the neutral condition, the participant chooses among existing voices in the supplied catalog; in the self-similar condition, recorded speech is used to create a synthetic voice resembling that participant. Both forms may produce familiarity or liking, but through different routes: preference is expressed directly when the neutral voice is chosen, whereas resemblance is introduced by synthesis and must be confirmed through participants' perceptions. For this reason, the manipulation is better described as a comparison between two forms of personalization than as personalized versus generic voice.

Research on customizing voice-assistant personality has also examined similarity between assistant and user [47]. Personality similarity, however, is conceptually different from acoustic self-resemblance: a listener can prefer a speaking style without hearing it as their own voice, and a close acoustic resemblance can be experienced as uncomfortable. The planned manipulation check therefore asks about perceived voice similarity, while the experience battery separately asks about liking, naturalness, familiarity, and uncanniness.

Voice choice has also been studied experimentally. In a programming game, Kao et al. independently varied whether participants could choose their avatar's visual appearance and voice [24]. Voice choice increased avatar identification and perceived autonomy when visual choice was also available, but did not produce those benefits when visual appearance was assigned. This conditional effect is relevant to the participant-selected neutral voice used here.

2.3 Self-Similar Voice and Embodied Collaboration

Two particularly relevant studies examine resemblance in different roles: an avatar speaking as the player and a virtual collaborator speaking to the participant. The present study extends this question to a guiding voice without a visible agent body.

Kao et al. analyzed 657 participants in an online programming game, comparing self-similar and assigned dissimilar avatar voices while also manipulating pitch [25]. Self-similar voice was associated with more puzzles completed, longer voluntary play, and higher similarity identification, competence, relatedness, and immersion. Direct differences on the reported intrinsic-motivation subscales were not significant.

The meaning of performance also differs across tasks. In Kao et al.'s game, longer voluntary play can indicate continued engagement, whereas in the present fixed search task a longer trial can indicate difficulty. Follow My Voice interprets completion time alongside workload, motivation, and reported reliance, without treating speed as a complete account of the value of self-similar voice.

Guo et al. used a 25-participant, within-subject design crossing appearance and voice self-similarity during virtual jigsaw collaboration [19]. Self-similar voices increased likability and believability ratings while also increasing eeriness. Voice did not significantly affect co-presence or perceived intelligence, although appearance affected those measures.

Both studies compared self-similar voices with assigned alternatives, whereas Follow My Voice compares the clone with a neutral voice selected by the participant [25, 19]. The question is therefore whether resemblance to oneself offers a different experience from a voice one has already chosen.

The present study also requires a distinction between the embodiment of the participant and that of the agent. Participants wear a headset and use a tracked controller, so their interaction involves physical sensing and action, but the guiding agent is not represented by a co-present animated character. Measures developed for an embodied collaborator are therefore reported as adaptations.

Applications of self-voice to daily reminders and emotional self-voice interventions extend this research beyond games and virtual collaboration [26, 16]. They motivate asking whether participants can imagine uses for their own synthetic voice. The debrief therefore begins with open questions before introducing probes about eeriness, self-talk, and possible applications.

2.4 Naturalness, Familiarity, and Uncanniness

Voice resemblance is not the only acoustic difference that listeners may notice. Synthesis can change articulation, prosody, loudness, duration, or perceived naturalness. Research on vocal uncanniness motivates measuring these qualities separately rather than subsuming them under similarity [12].

Both conditions use the same semantic library and speech provider. This controls content but not acoustics, because prepared clips can differ in duration and pronunciation even when their text matches. Pilot checks assess intelligibility and comfortable playback levels for neutral and cloned voices, and the recorded provider, model, voice, and library identifiers document the configuration each participant heard.

Custom experience items and the appealing, eerie, and familiar items selected from Guo et al. are analyzed separately, because no measurement model combining them has been established for this study. A participant may report a voice as both familiar and eerie, and preserving the individual item directions keeps that response interpretable.

2.5 Workload, Motivation, and Agent Perception

Task speed describes efficiency, whereas workload, motivation, and agent perception address different aspects of receiving help.

Zagermann et al. discuss eye tracking as a means of investigating cognitive load in visual computing [53]. This work motivates considering gaze-derived measures, but an unvalidated count of ray hits is not a workload measure. The confirmatory plan therefore relies on subjective workload dimensions and task performance; pupil, fixation, and saccade analyses remain contingent on separate acquisition and event validation.

Nerella et al. investigate gaze-based cognitive-load prediction during an AR color-matching task with interrupting alerts, using 31 participants and Magic Leap 2 eye tracking at 60 Hz [39]. Their group XGBoost model achieves a reported test AUC-ROC of 0.85. Positive labels are tied to participants' reports of overload in response to alerts, and the authors report no significant correlation between high-load report frequency and NASA-TLX. The study supports investigating gaze alongside subjective workload while preserving the distinction between momentary reports and retrospective ratings.

NASA-TLX supplies six workload dimensions [20]; this study adapts it with a 0–10 response range and an unweighted complete mean. The Intrinsic Motivation Inventory distinguishes interest/enjoyment from related subscales such as effort, pressure, and value [42, 35, 8], and the subscales are reported separately. The instrument appendix gives the item wording and reverse keys needed to reproduce scoring.

The agent-perception measures are taken from Guo et al., who attribute them to their original sources [19, 6, 36].

2.6 Visual Search and Attention Guidance in Extended Reality

To study a guiding voice, the task must give its instructions a meaningful role. Object search provides that role by requiring the participant to connect guidance with a location and then an intended object.

Search in a surrounding display differs from search on a single front-facing screen because relevant objects may lie outside the visible field. Turning changes what can be inspected, while eye movements refine attention within the visible region. Work on eye and head movements across an extended field of view is therefore relevant to the task geometry [48]. In the present study, this motivates recording both head motion and gaze, since neither fully substitutes for the other.

Several approaches to directing attention in immersive environments inform this design. Grogorick et al. address subtle gaze guidance, while HiveFive concerns immersion-preserving attention guidance [17, 30]. Ariza and colleagues investigate vibrotactile assistance and proximity-based multimodal feedback for selection [3, 2].

A recent paper, RippleVision, provides a visual counterpart to gaze-contingent spoken guidance. Kudnick et al. describe brightness ripples restricted to a cone between a point of interest and the current gaze location, with opposite brightness modulation in the two eyes [27]. This summary draws only on the published abstract; its detailed methods and comparative outcomes are not assessed here. The technique illustrates how assistance can adapt to a participant's gaze without identifying the target verbally. Follow My Voice holds the assistance policy constant across voice conditions and does not compare guidance modalities or assume that visual and spoken cues have equivalent effects.

Binetti et al. examine visual and auditory cues for locating out-of-view objects in head-mounted augmented reality [5], and Warden et al. focus on target cue type and location in augmented-reality

search [49]. Follow My Voice does not open with a color-and-shape description of the target; instead, the sequence of directions gradually narrows the participant's search.

The hidden-target design also distinguishes this study from known-target search. A participant who already knows the target can continue searching without further assistance, whereas here the participant must obtain enough spatial information to infer which object is intended. The voice may therefore be heard more often, and may matter more, than in a task where it supplements an already known target.

2.7 Eye Gaze and the Choice of Interaction Modalities

2.7.1 Gaze as Context for Assistance

Eye tracking provides an observation of where the user is looking that can be compared with the target's position. This makes it useful for adapting guidance as a search unfolds: a broad turn instruction may be relevant while the target is outside the current view, whereas a small vertical correction may be more useful once the user reaches its region.

Existing gaze-contingent auditory techniques provide precedents for this adaptation. Losing et al. vary a tone with gaze-to-target distance, and Kwok et al. combine initial orientation with finer guidance [33, 29]. Their desktop studies support investigating feedback tied to ongoing gaze. EmBARDiment offers a more recent XR example of using gaze context to support an assistant's interaction [7]. The present system applies this rationale to short spatial instructions while keeping the guidance policy common across voices.

2.7.2 Speech, Visual Feedback, and Deliberate Selection

The modalities have complementary roles. Gaze supplies context, speech communicates a direction, visual feedback helps users understand interaction state, and the right-controller trigger provides an explicit selection. Keeping selection separate from looking allows participants to inspect an object without automatically committing to it. Spoken directions also carry the voice-identity cues needed for the self-similar comparison, unlike a nonverbal tone or a visual arrow alone.

This division of roles motivates the apparatus; the study does not compare modalities experimentally. Both voice conditions use the same gaze and controller inputs, and the design includes no unguided control. Recognition and comprehension cannot be inferred from a valid gaze intersection alone; they require behavioral evidence and participant reports. Tracking validity and calibration also constrain when gaze can inform assistance, as discussed in the eye-tracking quality section below.

2.8 Gaze-Contingent Auditory Guidance

Auditory guidance takes several forms. A sound can encode gaze error through loudness, indicate a location through spatial rendering, or communicate an instruction through words. Only the last of these necessarily carries the verbal and vocal identity cues central to this thesis.

Losing et al. tested gaze-contingent sonification in a small desktop study with ten participants: the loudness of a flute tone varied with gaze-to-target distance, reaching silence at the target [33]. Benefits depended on the search task, with the clearest improvement in the difficult number-search condition. The study supports linking auditory feedback to current gaze but offers no evidence about any particular spoken phrase. The present adaptation replaces the continuously varying tone with short spatial instructions so that participants hear the assigned voice throughout search. The

frequency and timing of these instructions were set for this study rather than derived from the sonification work, and they have not yet been validated.

Kwok et al.'s Two-Step Gaze Guidance evaluated 69 participants using a three-monitor display and eye tracking, rather than a VR headset [29]. Initial spatial-audio orientation followed by finer guidance shortened search times relative to the other tested conditions; accuracy did not differ significantly across conditions. Their implementation also buffered transitions between guidance stages. The transferable idea is a staged change in the information a cue supplies: first orient toward a region, then refine within it. Follow My Voice adapts that structure to wall entry and within-wall directions.

Binetti et al. provide a complementary headset-based example using visual arrows and spatialized alarm sounds to locate out-of-view targets [5]. Adding sound mitigated response-time costs when visual cues and targets occupied different depth planes, but did not remove the corresponding accuracy cost. Because the study had no sound-only condition, it offers conditional support for an auditory channel when visual guidance is constrained rather than evidence that speech is superior to visual cues. Its separate response-time and accuracy results also motivate reporting selection errors separately from completion time in the present voice comparison.

Zhang et al.'s 96-participant AR book-search study more directly concerns verbal instructions [54]. Its audio hint specified an approximate shelf once at task onset; it was not repeated gaze-contingent speech. In the sub-study without gaze-playback feedback, hint conditions outperformed the control but did not significantly differ from one another. Advantages of combined visual and audio hints over either modality alone appeared in the second task of the feedback sub-study. These results support examining task completion and workload together and stating cue timing and feedback exposure explicitly.

Sabouni et al. evaluated nonverbal proximity alerts during a VR search for target paintings [43]. A brief beep signaled entry into a target's vicinity. Compared with no cue, the audio condition reduced task-list checking, but it did not significantly reduce completion time relative to the control. The study therefore supports examining attention allocation alongside completion time.

Coughlan et al.'s CamIO system guided a tracked stylus toward points of interest on physical objects using cardinal-direction instructions [11]. Its feasibility study involved four blind participants. This provides a precedent for communicating spatial corrections through speech.

Design synthesis for the voice comparison.

The agent first helps locate the relevant wall, then narrows the search within it; the participant confirms the inferred target with the right trigger. The target's initially hidden identity gives the voice an ongoing instructional role, unlike a single announcement supplementing a known target.

The design records guidance during search, acknowledgments of qualified gaze or controller contact, and feedback after selection as separate events. A success chime after selection cannot explain an earlier change of direction within the same trial, although it may shape later experience. The researcher-facing Trial Review tool also differs from Zhang et al.'s participant-facing gaze-playback intervention [54]: researchers use it to audit delivery, and the protocol includes no participant playback.

2.9 Context and Short-Term Memory in XR Agents

A related AR navigation system by Yang et al. combines Building Information Modeling with retrieval-augmented generation and compares directional arrows alone with an embodied guide

accompanied by the same arrows [50]. Its comparison concerns the presentation of navigation assistance, not self-similar voice. The walking task, destination queries, and embodied interface in that system differ from the seated hidden-target task used here.

EmBARDiment uses gaze-informed contextual memory to support interaction with an embodied XR agent [7]. That approach motivates making recent interaction evidence available to the assistance policy. Follow My Voice implements a narrower, bounded history of recent spatial progress, actual speech, and guidance transitions, with retention limits set for this application rather than taken from the cited system. The history holds no long-term autobiographical memory or inferred model of the participant's mental state. Unlike the cited interface, the guiding voice also has no visible agent body.

Work on memory-augmented AR agents and the GenLARP system illustrates further directions for context-aware XR agents [10, 52]. The former is a conceptual framework and the latter a short system description, so neither supplies a validated timing threshold for this application. Research on user-aware virtual agents and proactive AR assistance likewise informs the design of state-dependent behavior [18, 32].

Short-term history is useful when the same current geometry can have different meanings. A target to the left may be one the participant has not yet approached, or one they have just passed while turning right. A recently delivered cue can make another identical cue unnecessary. History distinguishes these cases only if it records reliable observations and actual playback. Because this mechanism is shared across voices, the study cannot estimate its independent effect.

Feedback specificity offers a further distinction between adaptive assistance and generic encouragement. Zimmerer et al. examined performance-specific feedback from a virtual exercise coach and found changes in selected agent-perception measures without a corresponding effect on motivation [55]. For Follow My Voice, this distinction motivates describing whether a cue reports

observed progress, corrects direction, acknowledges qualified gaze or pointing, or confirms a successful selection. These functions are common to both voice conditions, so the study compares voice configurations rather than estimating an independent effect of feedback specificity.

2.10 Eye-Tracking Quality and Interpretation

Eye-tracking quality involves more than a nominal sampling frequency. The reporting guidance of Dunn et al. and the data-quality discussion of Niehorster et al. motivate reporting acquisition, validity, spatial performance, and missingness [15, 40]. For this system, the relevant questions are how many distinct source observations arrive, how old they are when used, and how error varies across the search space.

Validation work on the Vive Pro Eye and tools such as GazeMetrics are methodological resources [45, 46, 1]. Their results do not transfer directly to the HTC Vive Focus Vision because device model, fit, calibration, coordinate conversion, and task geometry can differ. The validation protocol therefore uses independent targets across heights, boundaries, distances, and head orientations, kept separate from the targets used to estimate an offset correction.

Drewes et al. compare gaze during locomotion in virtual reality and the real world [13]. Because the present study is seated and relies on head and chair rotation rather than free walking, locomotion findings do not directly estimate gaze behavior in this task. Head motion is logged because it participates in search, and claims about eye events are limited to what the available measurements support.

Salvucci and Goldberg discuss methods for identifying fixations and saccades in eye-tracking data [44]. The 100-ms vicinity qualification used here governs interface acknowledgment and is

not a validated fixation classifier; gaze contacts and qualified vicinity events are therefore reported as interaction measures.

2.11 Comparison of the Closest Research Strands

Table 2.1 summarizes what each selected research strand contributes to this design and what it leaves unresolved. No single strand provides a baseline for the present study, because spatial guidance, voice resemblance, context management, and measurement quality address different questions.

Table 2.1: Selected research strands and their role in Follow My Voice. These are methodological connections, not evidence that the present implementation reproduces the cited effects.

Research strand	Relevance to this project	Boundary of transfer
Self-similar voice [25, 19]	Motivates the voice manipulation and both favorable and adverse experience measures.	Educational or embodied collaboration tasks do not establish a search-speed benefit here.
Auditory and staged gaze guidance [33, 29]	Supports investigating gaze-dependent, coarse-to-fine assistance.	Does not validate this headset, phrase policy, or gate durations.
AR search and task feedback [54]	Separates in-task cues from post-task feedback.	Known-target task and feedback manipulation differ from this hidden-target voice comparison.
Context-aware XR agents [7]	Motivates explicit recent context for state-dependent assistance.	Does not establish an independent benefit of this bounded memory.
Eye-tracking reporting and validation [15, 40, 1]	Informs source timing, validity, spatial validation, and gap reporting.	Reporting guidance and other-device results are not Focus Vision accuracy measurements.

In summary, the agent knows a target that is initially unnamed to the participant, the participant receives adaptive spatial speech, and explicit right-trigger selection remains separate from gaze. The experimental manipulation varies the voice configuration and holds this assistance mechanism constant.

CHAPTER 3: METHODOLOGY

This chapter describes the application, experimental design, procedure, and data collection used to examine self-similar spoken assistance in virtual reality. The study compares performance, perceived workload, motivation, and agent experience across self-similar and neutral voices within the same gaze-contingent search task. Voice condition is the experimental factor; gaze sensing, spoken spatial guidance, and right-controller selection remain common to both conditions.

The application runs on the HTC VIVE Focus Vision headset. Each participant completes one practice trial and 80 measured trials, divided into two counterbalanced blocks of 40 trials, one per voice condition. The project source is available at the Gaze Contingency Project GitHub repository.

The methodological choices are informed by Chiossi et al.'s visual-search study, which combines search outcomes with eye-tracking measures [9]. The present study adopts that separation between task performance and visual behavior while changing the research question to voice identity, withholding the target identity initially, and using a researcher-selected 25-second search limit. Complementary work by Pfeil et al. at UCF compares eye–head coordination in virtual and physical viewing [41]; it motivates retaining head motion alongside eye direction when interpreting gaze in this application.

3.1 Application Design

The application presents a virtual setup menu before the search task. Participants choose a neutral voice from the supplied options, record a standard passage for their self-similar voice, and confirm that both voices are clear and comfortable to hear. Voice preparation and eye-tracking setup precede

practice. During search, participants remain seated on a swivel chair, inspect the surrounding scene, and use only the right controller for pointing and selection.

3.1.1 Apparatus

The application is developed in Unity 6 (6000.3.10f1) using the Unity OpenXR Plugin (1.16.1), HTC VIVE OpenXR integration (2.5.1), and Unity XR Interaction Toolkit (3.3.1). It is deployed as a standalone Android ARM64 application on the HTC VIVE Focus Vision head-mounted display, with rendering performed locally on the headset using the Universal Render Pipeline (17.3.0). The study uses the headset's virtual-reality configuration, in which the headset supplies head-pose and eye-gaze tracking and the tracked right-hand controller supplies pointing and trigger input. Network access supports cloud-based agent decisions and voice preparation; it is not used to stream rendered frames from a host computer. HTC specifies 2448×2448 pixels per eye, a 90-Hz standalone display refresh rate, and a field of view of up to 120 degrees [22]. The advertised 120-Hz binocular gaze output is a sensor specification; the application's effective rendering and gaze-delivery rates will be measured on the study build.

3.1.2 Objects, Planes, and Selection

The search task takes place in a neutral virtual room. Seated participants interact using the right controller, while colored geometric objects hang on eight vertical planes arranged around them; Figure 3.1 shows the layout from above. Because the application knows the target identity but withholds it from the opening instructions, participants must rely on voice guidance to locate the target, which they select by pointing the controller ray at it and pressing the trigger. The stimulus generator includes six shapes—sphere, cube, pyramid, cylinder, capsule, and star—in red, blue, yellow, and green. Eight vertical planes surround the seated origin at 45-degree intervals. The development

configuration contains 168 objects, or 21 per plane, at a default plane radius of 1.5 m from the seated origin; these object counts are unrelated to the eight trials assigned to each absolute nominal angle in a 40-trial block. Objects vary in height and horizontal position within their assigned plane, so search requires both coarse orientation toward a plane and finer search within it. For each trial, the system logs object positions, layout seed, wall assignment, and target identity.

Each wall is 2.2036 m high (approximately 2.20 m), 30% lower than the 3.148-m walls of earlier development layouts. Object centers occupy heights from 0.10 m to 2.1036 m above the floor, leaving a 0.10-m margin above the highest permitted center, and are separated by a minimum three-dimensional center-to-center distance of 0.20 m. Layouts are generated from a saved seed; for measured trials, final positions also depend on the eye origin saved at readiness completion.

Both voice conditions use the same layout and spatial constraints. Pilot sessions will assess seated comfort, visibility, ray selection, and realized height allocation before these parameters are fixed for participant collection.

Figure 3.2 complements the schematic with a reconstruction from an actual headset development recording. It shows the first-person view, surrounding layout, and recorded shape types. The recording predates the October 23 wall-height reduction, so its 168-object layout uses the earlier, taller wall geometry.

3.1.3 Voice Preparation and Audio

All voice audio is pre-synthesized with ElevenLabs, which supplies both the neutral voice and the self-similar voice. To support the self-similar condition, each participant records the same standard passage in their usual speaking voice. Both conditions share the semantic phrase library, guidance policy, and playback routing, so voice is the intended manipulation. Because guidance

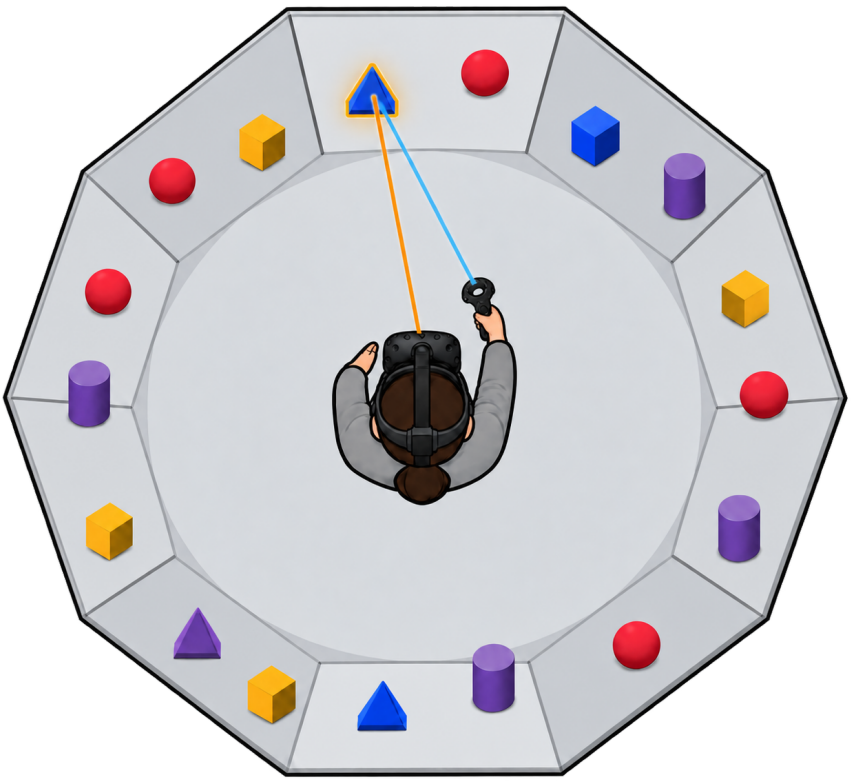


Figure 3.1: Top-down view of the search scene. The seated participant is surrounded by wall panels holding colored geometric objects. The gaze ray (orange) and controller ray (blue) converge on the highlighted target, the state in which the participant is ready to select with the trigger. The layout and drawn object count are schematic; object positions and target identity vary by trial. The converging rays and highlighted endpoint illustrate the selection interaction, not a cue that reveals the target before search.

responds to behavior, however, the wording and number of hints can differ between blocks, and the synthesized clips can differ acoustically. The preparation pipeline automatically checks wording and library readiness, but these checks do not assess perceptual equivalence; pilot sessions will therefore verify per-voice intelligibility, loudness, and acoustic comparability. No pitch or spatial effects are applied to the self-similar voice. The system logs the provider, model, voice ID, library version, and synthesis settings.

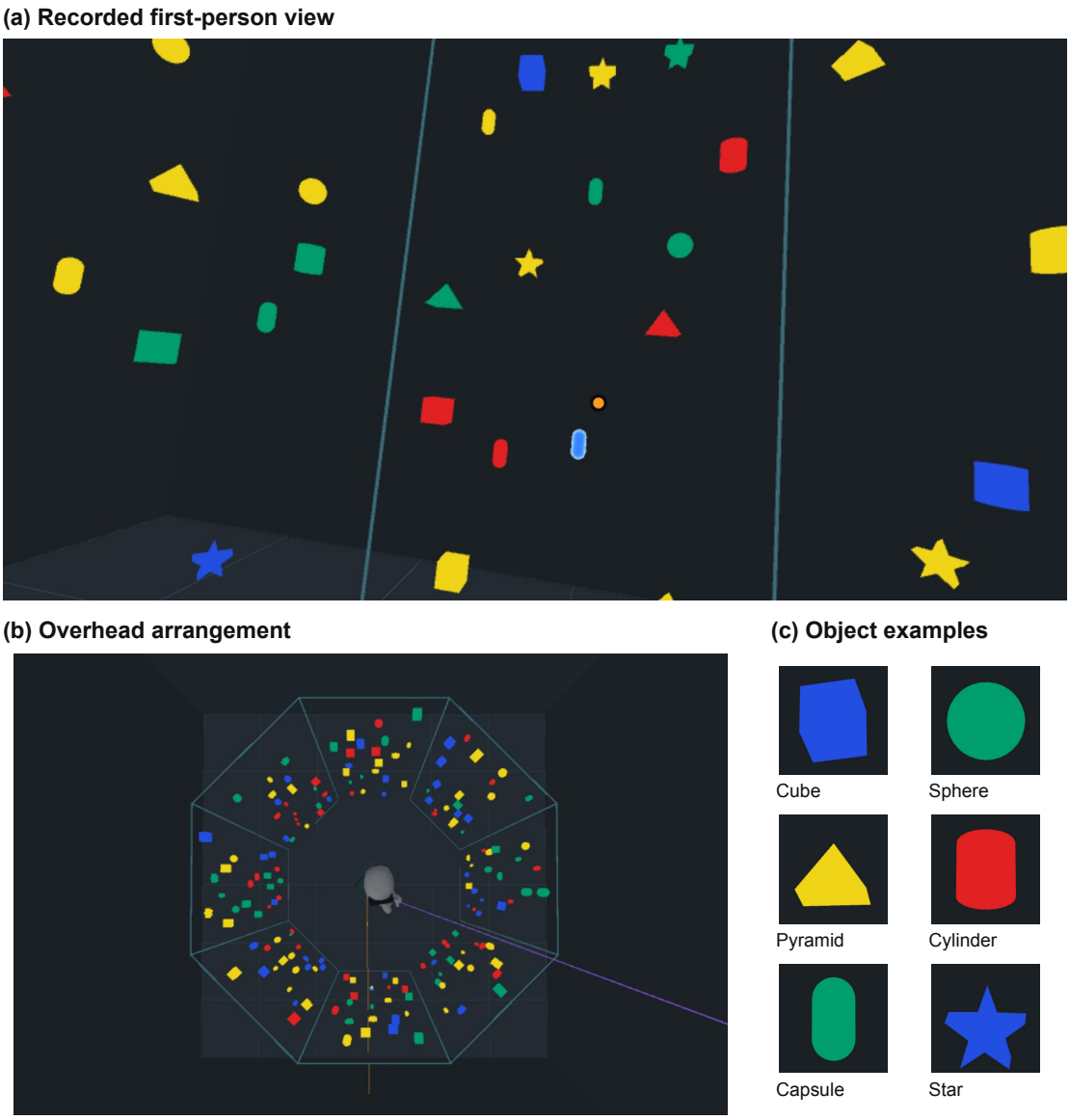


Figure 3.2: (a) First-person view of colored objects on surrounding vertical planes. (b) Overhead view of the eight-plane arrangement. (c) Enlarged examples of the six shape types at independent scales. Images are reconstructed from a headset development session recorded before the wall-height reduction and show the earlier wall dimensions; they are development material, not participant-study results.

The Trial Review application aligns gaze and controller state, guidance transitions, agent decisions, and playback events with trial geometry on a common timeline. Trial-level aggregates are stored separately from the raw gaze, motion, and event streams. The effective gaze sampling rate will be determined from distinct source timestamps on the study headset build rather than from the nominal specification or the application polling rate. The analysis will report ray intersections as gaze hits, which approximate, but are not equivalent to, fixations identified by a validated event classifier.

3.1.4 Sensing and Guidance State

The system updates spoken directions from evidence of the participant's progress, so that instructions can shift from orientation toward a wall to refinement around an object. The same sensing and state logic serves both voice conditions.

The system continuously collects eye-gaze origin, direction, and validity; head pose; right-controller pose; and ray-object intersections. A local state machine combines these observations with the target geometry to track the participant's progress through the search: turning toward the target wall, refining within the wall, acquiring the target with gaze, pointing with the controller, and becoming ready to select. Additional states cover suspended or paused operation. Each state transition is logged with its cause.

The state machine issues new gaze-dependent guidance only when gaze data are valid and recent. Gaze ray hits, gaze-vicinity candidates, and controller hits are treated as separate observations; when gaze-vicinity regions overlap, the closest object is chosen as the vicinity candidate. A controller ray hit on the target immediately suppresses contradictory directional cues, whereas the spoken pointing acknowledgment separately requires a one-second controller hold with valid nearby gaze when the speech scheduler permits the acknowledgment to play. Gaze confirmation, controller confirmation, and trigger selection are recorded as distinct events.

3.1.5 Agent Decisions and Playback Control

Jev, the cloud decision agent shared by both voice conditions, selects from a constrained set of options rather than generating free-form instructions. Local geometry and state rules first filter the phrase library to candidates that are valid in the current situation, and each request to Jev then contains a state snapshot, recent memory, playback status, and two to five eligible phrases. From these, Jev chooses the wording and whether to speak now or wait, while target definition and geometric reasoning remain local. Eligibility filtering and playback revalidation are intended to reduce spatially incorrect or stale guidance.

Because cloud responses arrive asynchronously, the participant's state may change before a decision returns. A local speech scheduler therefore revalidates each decision against the current state, evidence, pacing, and object intersections before playback, and discards decisions that are stale or conflict with the current situation. Qualified gaze and controller acknowledgments, together with eligible turn-progress milestones, can be handled locally and played from prepared phrases without waiting for a new cloud decision; they remain subject to the same checks. When a qualified target acknowledgment would replace misleading search speech, a local fast path avoids waiting for interruption advice. This path protects the first 0.35 seconds and final 0.5 seconds of the current phrase, retains a 2.5-second interruption cooldown, and fades the audio over 150 ms. The fast path keeps the separate gaze and controller hold requirements, and fresh evidence is checked before the fade, after it, and at playback. Setup, completion feedback, and existing acknowledgments are protected from interruption. Each completed or interrupted utterance updates both the memory and the event log. Model latency, deferral, cancellation, fallback, and playback onset are logged separately to support delay attribution. When a trace lacks the required events, the source of a delay is recorded as unknown rather than inferred from the spoken phrase alone.

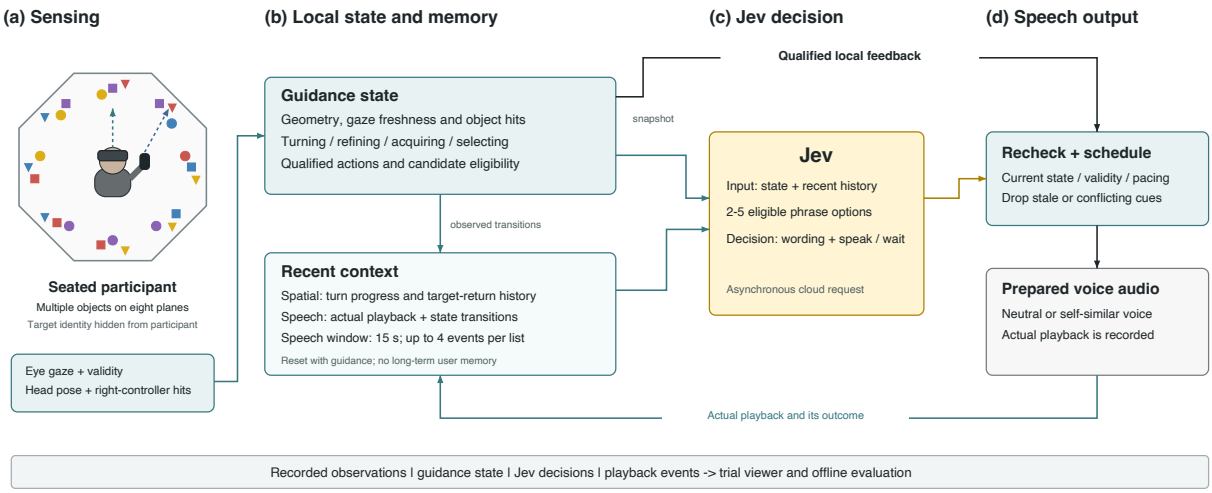


Figure 3.3: System architecture. (a) Gaze, head, and controller data are sensed. (b) A local state machine and short-term memory determine which guidance actions are valid. (c) Jev chooses wording and timing from eligible options. (d) The scheduler revalidates each cue against the current state before playback. Actual playback feeds back into memory. The scene inset is schematic and does not reflect stimulus count or any participant-visible target cue.

Figure 3.4 shows the final playback gate. Because Jev's response is based on an earlier snapshot, it is treated as advice: a decision to speak results in playback only if the cue passes current-state checks. Provider-selected phrases and qualified local feedback both pass these checks before the prepared clip starts. Readiness and pacing are checked during scheduling, and the queued relevance callback is evaluated again immediately before audio playback. A development-only Jev-led policy broadens the choice to eligible orientation, refinement, correction, or encouragement actions while retaining the same local gate.

3.1.6 Trial Review Interface

Figure 3.5 shows the standalone Trial Review application with its bundled synthetic demonstration. The scene and inspector share a timeline with speech, gaze, events, and decision tracks. The appli-

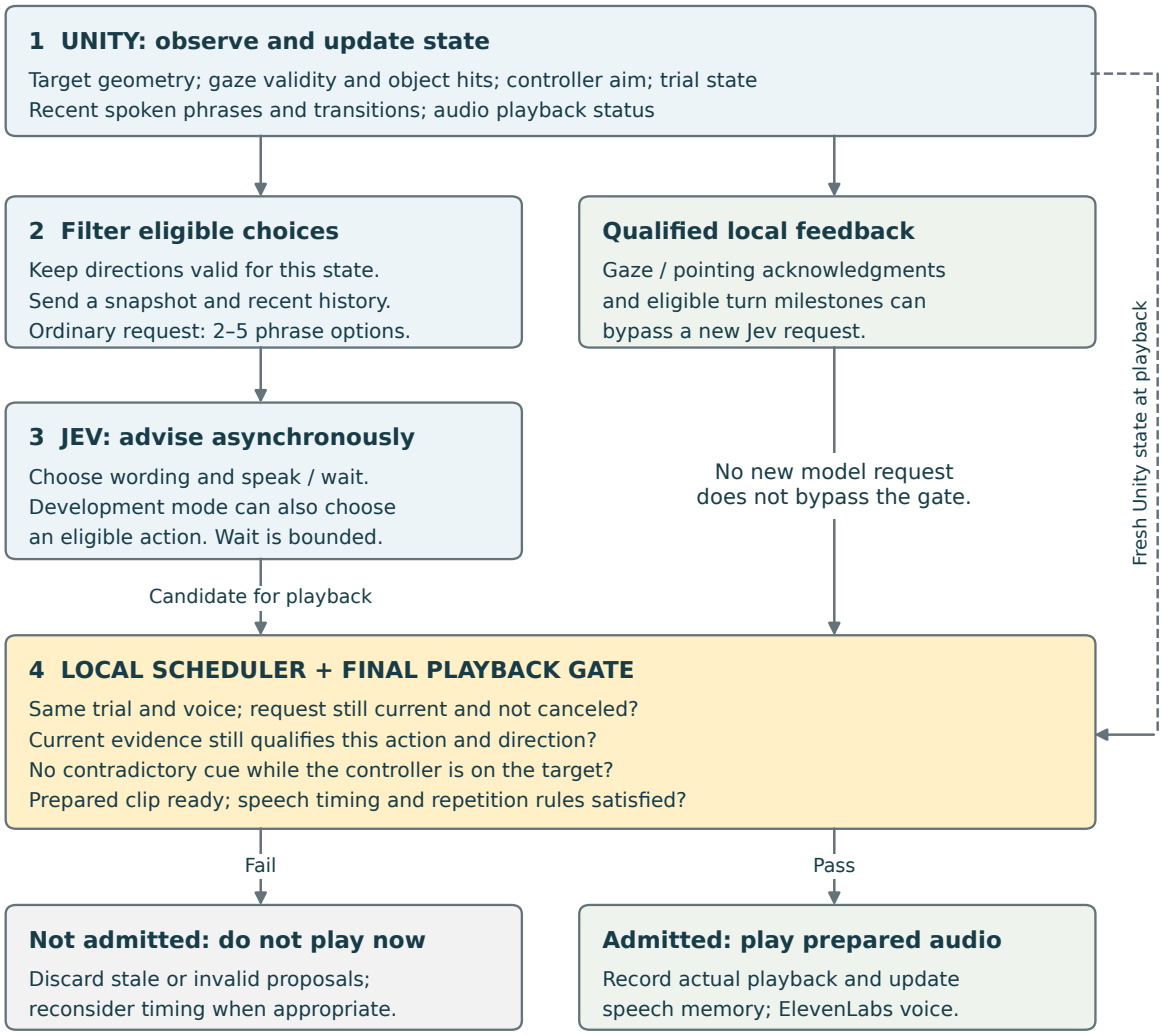


Figure 3.4: Jev decisions and the final local playback gate. The dashed input indicates that the gate evaluates fresh Unity state, not only the snapshot originally sent to the provider. Box 4 groups scheduling checks with the final relevance check immediately before playback. Qualified local feedback bypasses a new model request but retains this boundary. Waiting is bounded by local policy; timeout and deferral fallbacks also pass through the gate. This schematic summarizes ordinary guidance and local feedback rather than every interruption, opening, or success-feedback path. It describes control flow, not measured response latency.

cation can also receive headset records for live review; transfer and receipt times are kept separate from source event times, so late arrival in the viewer does not indicate delayed guidance on the headset. The capture illustrates the interface, not a participant recording or performance result, and the demonstration explicitly marks missing state history and reconstructed highlights.



Figure 3.5: Application screenshot of the bundled SYNTHETIC / Run 1 demonstration. The central reconstruction, right-hand inspector, and timeline provide synchronized review. Missing state, estimated rings, and reconstructed highlights are labeled by the application. This is a visualization example, not headset video or empirical study data.

3.1.7 Delivery Quality as an Experimental Requirement

Both voice conditions use the same versioned guidance mechanism, although delivered sequences depend on participant behavior and clips may differ acoustically. Voice identity, library metadata, playback onset, and delivery faults are therefore recorded so that each can be traced to its condition.

A delayed or confusing cue could alter search performance or a participant's judgment of the voice, and could then be mistaken for an effect of voice condition. Appendix C defines guidance sequence tests and an evaluation rubric for checking the implementation.

3.2 Experimental Design

Draft status. This section specifies the proposed confirmatory procedure. The implemented snapshot supports one practice trial, two 40-trial blocks, and a final questionnaire handoff. It also records pre-cue starting geometry and balances target direction and height within each block. Sample size and pilot acceptance rules remain unresolved.

3.2.1 Voice Manipulation and Comparison Rationale

The method compares two ways of personalizing assistance: a synthetic voice derived from the participant's recorded speech and a neutral voice selected by that participant from the supplied catalog. Letting participants choose the neutral alternative ties the comparison to a design choice a user could face, while the within-participant task provides repeated observations under both configurations. Similarity ratings indicate whether participants heard the clone as resembling themselves, whereas performance and experience measures compare how each voice functions as guidance.

Gaze contingency and the agent's decision rules are common to both conditions. Because the target identity is initially hidden, participants receive repeated guidance in the assigned voice on every trial. Jev, memory, and gaze awareness are held constant rather than varied as independent factors.

3.2.2 Design and Trial Allocation

The study uses a within-participant comparison of two voice conditions: a neutral voice and a self-similar synthetic voice. Both conditions use the same guidance policy, semantic phrase library, selection method and visual-feedback settings, although actual phrase sequences may differ because guidance responds to behavior. Because the neutral voice is chosen by the participant whereas the clone is derived from the participant's speech, the two voices may also differ in preference, gender presentation and synthesis quality; the contrast therefore cannot isolate self-similarity from these features.

A saved, balanced randomized allocation assigns neutral-first or self-similar-first order. Each participant completes one practice trial followed by two measured blocks of 40 trials, one per voice. The practice trial uses the first voice in the assigned order (the neutral voice for Order A, or the self-similar voice for Order B) and is excluded from measured outcomes. A seed makes the trial schedule reproducible. Nominal horizontal-angle quotas are identical across blocks, with trial order and left/right assignments randomized subject to those quotas. Within each block, each bilateral angle category (45, 90 and 135 degrees) has four left and four right assignments.

Because objects vary in position within each plane of the shared layout described above, each trial specifies the exact starting angle and target height in addition to the assigned wall. Vertical allocation is defined by target-center height relative to the readiness cross rather than the participant's seated eye height. The cross is placed at the midpoint of the permitted object-height range, and each assigned target center lies between 10 and 45 percent of that range above or below it. The lower

Table 3.1: Study allocation. Nominal wall angles are schedule categories, not exact object bearings or accumulated head rotation.

Parameter	Allocation
Voice conditions	Neutral voice; self-similar synthetic voice
Measured trials	40 per block; 80 per participant
Practice	One trial; excluded from outcomes
Absolute nominal wall step	Eight trials each at 0, 45, 90, 135 and 180 degrees per block
Vertical allocation	20 targets above and 20 below readiness-cross height per block; four above and four below at each nominal angle
Condition order	Counterbalanced neutral-first and self-similar-first
Active search limit	25 seconds, excluding explicit pauses
Selection	Right-controller ray and a trigger press

bound keeps every target clearly above or below the cross, and the upper bound keeps targets away from the floor and the top of the wall; pilot sessions will assess target visibility and seated comfort at both extremes.

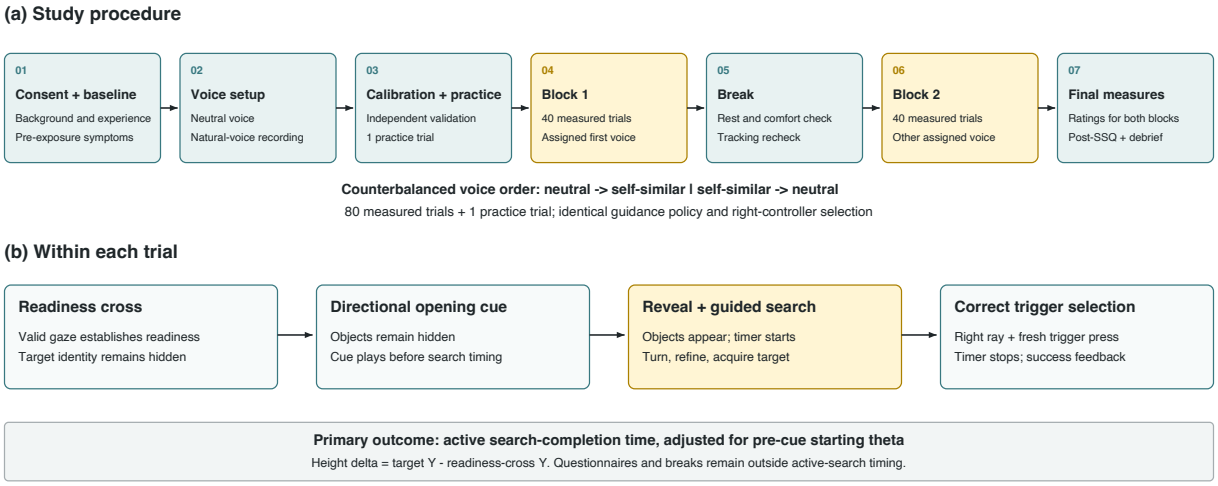


Figure 3.6: Study procedure and trial timing. Opening guidance precedes object reveal; active completion time ends at correct right-trigger selection. The success path shown here is bounded by a 25-second active-search deadline; expiry ends the attempt without a successful completion time. Questionnaires and between-trial breaks remain outside active-search timing.

Nominal angle steps are defined relative to the readiness (start) wall, which is the previous target wall. For the first measured trial, this reference is the practice target wall. The first measured cross is centered on its wall; each later cross is shifted sideways by a seeded offset of up to 20 percent of the usable half-width. When readiness completes, while all objects remain hidden and before the opening cue, the application rotates the horizontal bearing from the participant's eye origin to the cross by the assigned signed angle, places the target where that bearing meets the assigned wall, and arranges the remaining objects around it. The exact starting angle defined below therefore equals the assigned category at that instant. If the required position falls outside the wall, the session stops so the participant can recenter, rather than substituting a different angle or trial. Because final positions depend on the saved eye origin as well as the seed, each layout is reconstructed from the saved object positions.

3.2.3 Participants

Eligible participants are consenting adults able to understand the task, hear the guidance, view the scene with usual correction and safely use the seated setup and right controller. Handedness, visual correction and prior XR and voice-assistant experience are recorded. Sample size and the enrollment ceiling will be determined before recruitment by a power analysis for the theta-adjusted repeated-measures outcome that accounts for plausible within-participant variation and incomplete observations.

3.2.4 Research Questions

The primary question is whether self-similar and neutral voices differ in active completion time after accounting for absolute starting theta and repeated observations within participants. Secondary questions concern workload and the four IMI subscales, while exploratory questions concern agent

perception, eeriness, voice experience, reliance, search behavior and variation in the voice contrast with starting angle. The primary voice contrast is two-sided because no directional advantage for self-similarity is assumed. Repeated trials are treated as within-participant observations, not as independent participants.

3.3 Procedure

Participants perform an agent-guided object-search task while seated on a swivel chair, with multiple colored geometric objects scattered across eight surrounding vertical planes. Although the application knows the target identity, the participant does not receive a color-and-shape description before searching. Instead, the agent guides the participant toward the target wall and then refines the direction horizontally, vertically or diagonally. The participant selects an object using the right-controller ray and trigger; looking at an object, receiving a gaze confirmation or pointing at it does not itself complete the trial.

Each trial begins with the participant looking at a readiness cross. Once readiness is established, an opening directional cue plays while the objects remain hidden, and object reveal then starts active search timing. A correct trigger selection before the 25-second active-search deadline stops timing and produces success feedback; an incorrect selection is counted and search continues within that same limit.

If the target has not been selected after 25 seconds of active search, the trial ends as a timeout. The application hides the objects, cancels pending spatial hints, and plays a prepared encouragement in the assigned voice, beginning with "Time up," before presenting the next readiness cross. For example, the agent may say, "Time up. Let's shake that round off and start the next." Separate wording introduces the block break or acknowledges the final trial. The same deadline and phrase

set apply to both voice conditions and to the single practice trial; a timeout advances the scheduled attempt without adding a replacement. Explicit pauses suspend the clock, whereas tracking loss alone does not. The 25-second limit is a researcher-selected protocol parameter, not a threshold drawn from the cited studies.

Before participant use, consent materials will be updated to describe ElevenLabs voice processing and Jev cloud decisions, including the data transmitted, retention, access, and deletion arrangements. After consent, participants complete background questions. They then choose a neutral voice, rate auditioned voices, record a standard passage in their natural voice for synthesis, and confirm comfortable, clear audio for each condition. The English recording passage is reproduced in Appendix A; its correspondence to the study build and its typical recording duration have not yet been verified. The neutral voice is chosen explicitly rather than inferred from participant gender.

The researcher fits the headset and conducts vendor calibration followed by independent spatial validation across relevant heights, wall boundaries, distances and head orientations. If an offset correction is introduced, the targets used to fit it are kept separate from those used for independent validation. Standardized instructions explain that participants rely on the agent to discover the target and use the right trigger to select it. Participants then complete the single practice trial and the two assigned 40-trial blocks.

No questionnaire interrupts the block sequence. Instead, during the final survey the same condition-specific questionnaire is completed twice, once referring to Block 1 and once to Block 2, with identical item wording and order. Breaks are offered between trials and at the block boundary, and any necessary in-trial pause is logged immediately. Tracking is rechecked between blocks and after headset adjustment. After headset removal, participants complete the two retrospective condition-specific sets, comparative voice preference and debrief. Participants may request a pause or stop, and symptoms or technical problems prompt researcher assessment.

Figure 3.7 distinguishes administration time from reference scope: both condition-specific batteries are completed in the final session, but each response refers to a single measured block.

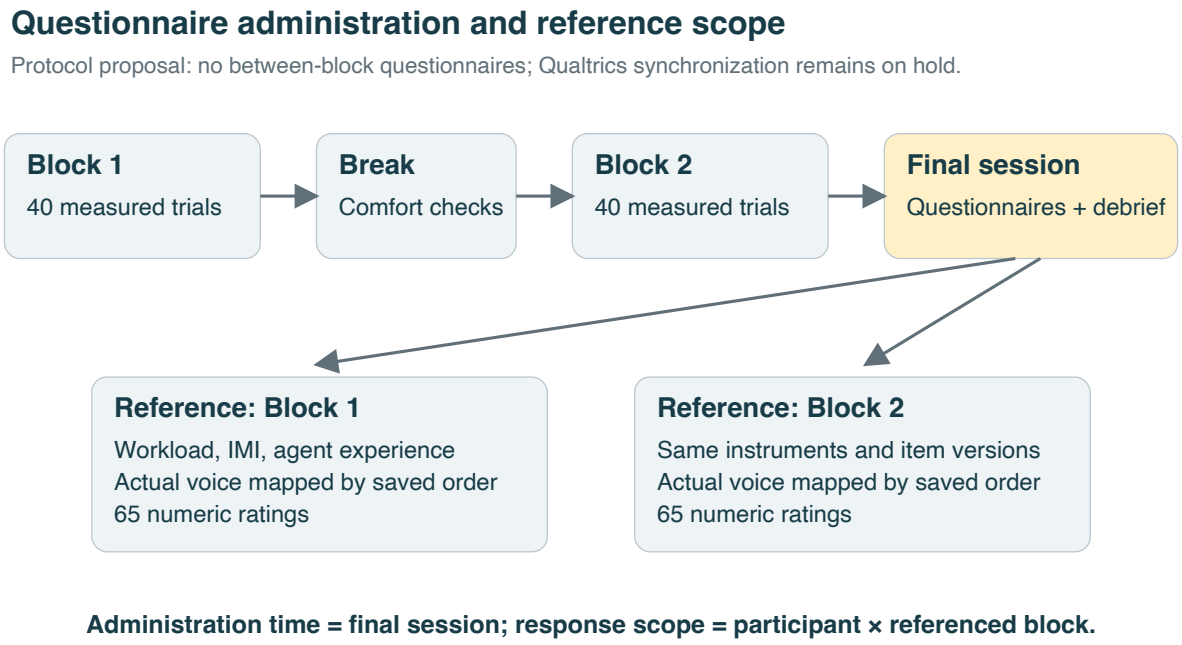


Figure 3.7: Proposed final retrospective administration. The intervening break contains comfort and tracking checks, not a block questionnaire. Each block-referenced battery contains 65 numeric ratings. Actual voice is recovered from the saved counterbalanced order.

The final survey will include an unambiguous reminder linking each block reference to the voice heard in that block. The reminder format (such as neutral labels or replayed samples) has not yet been decided and will be piloted to check that it adds no differential cues between conditions.

3.3.1 Pilot Testing and Parameter Checks

Before confirmatory recruitment, pilot sessions verify the realized angle/height schedules, target visibility and seated comfort, readiness and selection behavior, gaze validity, audio timing and

wording, and reconstruction of trial events from the saved data. Pilot assessment also measures survey comprehension and completion time. These checks inform calibration-acceptance, practice-proficiency and technical-stop criteria. The pilot also assesses whether the 25-second deadline is suitable, and the outcome is documented with the study build and questionnaire versions.

3.4 Data Collection

Measuring the voice comparison involves three questions: whether participants perceived the self-similar voice as resembling their own, whether voice condition changed search performance, and how participants experienced the assistance. Answering them requires outcome measures together with an account of the speech each participant actually heard. This section therefore links voice identity, delivered speech, search events, and questionnaire responses, so that perceived similarity can be reported as a manipulation check and kept distinct from performance and experience outcomes. Detailed gaze and timing records support interpretation of these outcomes, and Appendix A retains instrument wording and scoring.

3.4.1 Frame-Level Records and the Trial CSV

The recording separates time-varying observations from trial summaries. Frame records retain application time, gaze origin and direction with tracking validity, head pose, available controller poses and input state, and the gaze/controller object-intersection evidence used by guidance. Object identities and transforms provide the scene needed for reconstruction; guidance transitions, agent decisions, actual speech playback, trigger selections, and pauses are recorded as events. Because a rendered frame can reuse a sensor observation, frame counts differ from the number of independent eye-tracking samples.

The trial replay snapshot stores the wall bottom in recording-local coordinates, the wall height in metres, and a flag indicating that these bounds are available; Trial Review uses them to reconstruct the walls. Earlier recordings, which lack these bounds, are reconstructed from object-derived wall estimates and their original object positions, without rescaling to the current layout. These replay fields supplement the saved layout version and transforms and are separate from the trial CSV columns. Because wall geometry differs between the two layouts, analyses that combine recordings from both retain the layout version.

Trial Review generates an Excel-compatible trial CSV with one row per trial. Its fields describe task performance, geometry, guidance exposure, and associated subjective responses (Table 3.2). Outcome, practice, provenance, and quality information accompany the exported records to support analysis and eligibility checks.

Table 3.2: Contents of the trial CSV. Unavailable values and reserved, uncollected fields are left blank.

Group	Recorded or derived columns
Identity	Participant ID, trial ID, block index, voice condition.
Geometry	Absolute and signed nominal theta; actual signed head-to-target yaw; target-minus-head height; target distance.
Guidance and duration	Instruction count and search exposure in seconds.
Target contact	Time to first and final target gaze contact; first right-controller target contact; final selection.
Subjective context	NASA-TLX score; reserved subjective performance rating, left blank pending an instrument mapping.

The export's yaw and height are head-relative and measured after onset; the pre-cue, cross-relative geometry is stored in the trial summaries and event log. Gaze-contact times are ray intersections rather than validated fixations, and a questionnaire score repeated across trial rows remains a single block-level observation.

3.4.2 *From Interaction Events to Research Variables*

The recorded data serve two purposes: reconstructing what happened in a trial and estimating the study outcomes. Reconstruction requires observations and events at their original temporal resolution, whereas statistical analysis usually requires a more compact table. The project therefore keeps trial summaries separate from gaze, head/controller motion, object layouts, and event streams, treating each summary as a derived representation rather than a replacement for the underlying evidence.

Each derived value has an identifiable source and definition. A number called "time to gaze" is ambiguous unless it specifies the clock origin, the contact rule, validity handling, and the treatment of missing observations; the recorded data keep these distinctions in separate fields.

Figure 3.8 shows how a trial CSV remains linked to the underlying observations, so that values prepared for statistical analysis can still be traced to the records needed for trial reconstruction.

3.4.3 *Coordinate Frames and Starting Geometry*

Let $\mathbf{o}$ denote the saved trial reference origin, $\mathbf{c}$ the readiness-cross center, and $\mathbf{p}$ the target center. All three positions are expressed in the same world coordinate frame. Let $\Pi(\mathbf{v}) = (v_x, 0, v_z)$ project a vector onto the horizontal plane. Define

$$\mathbf{a} = \Pi(\mathbf{c} - \mathbf{o}), \qquad \mathbf{b} = \Pi(\mathbf{p} - \mathbf{o}). \tag{3.1}$$

For nondegenerate vectors, the exact pre-cue starting angle is

$$\theta_0 = \operatorname{atan2}(\hat{\mathbf{y}} \cdot (\mathbf{a} \times \mathbf{b}),\ \mathbf{a} \cdot \mathbf{b})\ \frac{180}{\pi}. \tag{3.2}$$

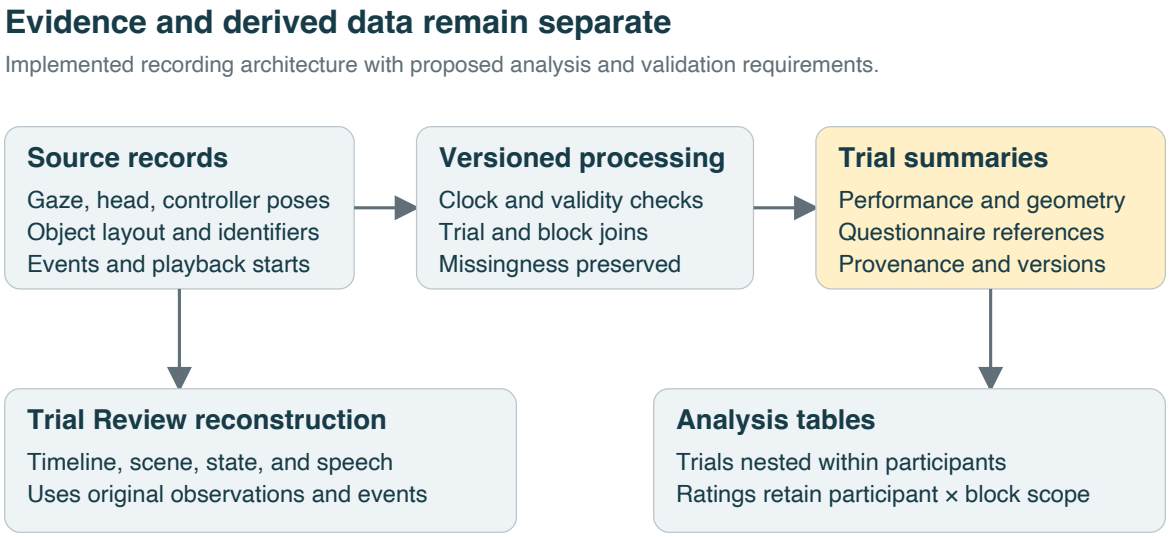


Figure 3.8: Recording and processing relationships. Source streams support reconstruction; derived summaries support analysis. Processing and scoring follow versioned definitions. Block-level ratings repeated in trial rows remain single block-level observations.

The sign convention is checked against Unity's world-up signed-angle convention using known left and right targets. The absolute value $|\theta_0|$ describes horizontal demand from the saved cross reference. For exactly backward targets, -180 and $+180$ identify the same horizontal bearing, so the sign depends on a tie convention while the absolute magnitude is 180 degrees in either case.

The application saves this snapshot when readiness completes, before the opening cue. It records the eye origin of the valid gaze observation that completed readiness, the cross and target centers in world coordinates, that observation's sequence number and receipt time, the assigned and realized signed angles, and the target-height difference. Because the target is placed from the same origin, the realized angle equals the assigned category; a difference larger than 0.01 degrees stops the

session. Recordings made before the snapshot was added, including the development recordings in Section 4.1, contain only nominal signed and absolute theta fields, which describe wall geometry relative to the previous target wall rather than the exact pre-cue angle. If a horizontal vector is degenerate, the angle is recorded as unavailable rather than zero, since zero would describe an easy starting configuration.

Each block contains eight scheduled trials in each absolute category, $0, 45, 90, 135, 180$ degrees. The target is generally offset from the center of its assigned wall, because its position is chosen to realize the assigned angle from the saved cross reference. Matching the starting angle controls horizontal demand only at that instant; total head and eye movement, vertical demand, and local clutter can still differ between trials, and movement during the opening cue and search is treated as an outcome. Both assigned and realized geometry are inspected.

The signed height difference is

$$\Delta h = p_y - c_y. \tag{3.3}$$

Positive values identify targets above the cross center and negative values identify targets below it; the reference is the cross center rather than seated eye height. Each block assigns 20 targets above and 20 below the cross, four of each at every nominal angle, with assigned target centers between 10 and 45 percent of the permitted object-height range from the cross, so no target is scheduled at cross height. The snapshot records the realized height difference, which is compared with the assigned sign.

The three-dimensional distance measure relies on a different observation: the first valid tracked head position $\mathbf{h}_k$ after reveal. Its value is $d = \|\mathbf{p} - \mathbf{h}_k\|_2$, in metres, stored with the timestamp of $\mathbf{h}_k$. Because it describes geometry at an observed post-reveal head pose, it cannot stand in for the pre-cue snapshot in recordings that lack one.

Figures 3.9 and 3.10 illustrate these definitions with conceptual rather than recorded layouts.

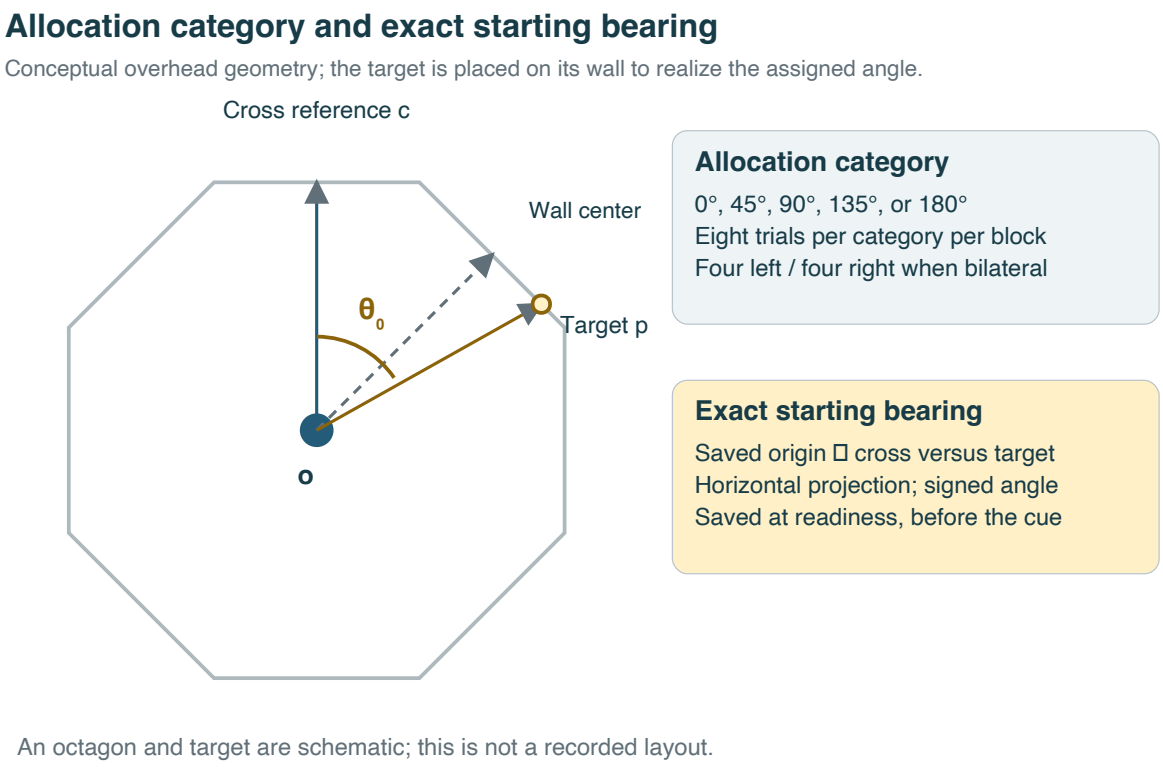


Figure 3.9: Conceptual overhead geometry. The target is placed on its assigned wall where the assigned angle from the origin-to-cross bearing meets the wall, so it is generally offset from the wall center. The pre-cue snapshot saves the origin, cross and target used for this placement.

3.4.4 Trial Clocks and Pause Intervals

Let t_r be object reveal, t_s correct selection, and $\mathcal{P}$ the union of valid pause intervals. Active completion time is

$$T_{\text{active}} = (t_s - t_r) - \mu\big([t_r, t_s] \cap \mathcal{P}\big)\,, \tag{3.4}$$

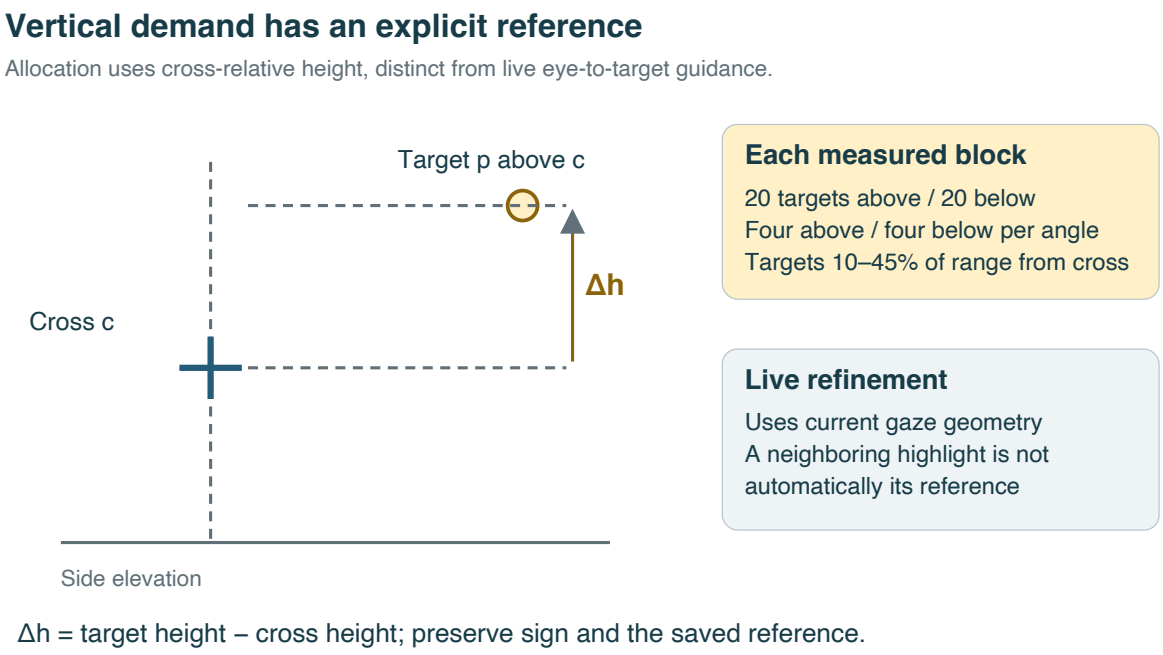


Figure 3.10: Conceptual side elevation for the signed height difference. The target shown above the cross has positive height delta. Each block balances targets above and below the cross; live vertical guidance uses current gaze geometry.

where μ denotes elapsed duration. Using the union prevents overlapping pause reasons, such as an application-focus pause that coincides with a researcher-requested pause, from being subtracted twice. The active clock remains stopped until every relevant pause reason has been resolved.

Readiness and opening speech precede reveal and are deliberately excluded from active search time. The broader interval from opening-cue onset to selection is also defined. It includes the opening instruction, whose duration may differ across voices; active completion time remains the primary outcome.

A stopped or timed-out trial has an observed duration and an outcome reason but no successful completion time. A timeout records 25 seconds of active search exposure, a distinct timeout event, and an unsuccessful outcome. The software detects expiry during an update and caps the exposure

clock at the deadline, while event timestamps retain the actual detection time. Earlier development recordings retain their original durations and are not retrospectively truncated to this limit. Source and receipt timestamps are likewise kept distinct, because a file that arrives late in the viewer may reflect transfer delay rather than a delay in the headset event.

Madritsch et al. distinguish overall task-completion time from active working time in an assistant-supported procedural task [34]; the present active clock is defined independently of their measure. In Follow My Voice, active completion time includes ordinary waits for guidance between object reveal and correct selection, and only recorded pauses are excluded. Separate decision and playback timestamps help explain delivery delays without removing their task-level consequences from the primary outcome.

3.4.5 Missed Targets

A missed target is a search that reaches its recorded 25-second active-search deadline without a correct trigger selection. An incorrect selection followed by a correct selection before expiry is a completed search, with its selection error retained separately. Technical stops, withdrawal, and recordings without a known terminal outcome are not automatically classified as misses. For each participant, block, voice condition, and recorded deadline, the descriptive missed-target percentage is

$$M = 100\frac{N_{\text{timeout}}}{N_{\text{selected}} + N_{\text{timeout}}}. \tag{3.5}$$

The denominator excludes practice and unresolved or technically interrupted attempts; their counts are reported alongside the percentage. A zero denominator yields an unavailable percentage. In the full trial export, `missed_target` is 1 for timeout, 0 for successful completion, and blank otherwise. A separate `missed_targets.csv` records grouped counts, deadlines, and development-data

exclusion labels. Legacy records without an explicit deadline are retained but excluded from this deadline-specific percentage.

This outcome follows Chiossi et al.'s distinction between successful search and searches ending without selection [9]. Because their task used a five-second limit and the present task is verbally guided with a 25-second limit, percentages are interpreted within each protocol rather than compared as benchmarks. The six unfinished searches in the earlier development recordings are reported as unfinished rather than as missed targets.

3.4.6 Eye-Event and Pupil Measures

An offline analysis preserves per-eye directions and pupil diameters from the HTC query stream, deduplicates repeated query timestamps, and separates search from readiness and pauses. Query receipt times record when software obtained the data; sensor capture time and the independence of consecutive observations remain unestablished. Changes in trial or device, invalid tracking, reversed timestamps, and query gaps longer than 25 ms break continuity. Invalid per-eye observations and observed eye closure exclude nearby 200-ms intervals.

The fixation detector uses elapsed-time dispersion: the sum of horizontal and vertical angular ranges must remain within 1.8 degrees for at least 83 ms. This starting configuration follows the I-DT parameters reported by Chiossi et al. [9], within the detector family described by Salvucci and Goldberg [44]. Azimuth is unwrapped across the signed-angle boundary. The implementation retains event intervals and the duration of eligible segments; its counts are candidate classifications rather than validated fixations.

The saccade detector operates separately on eye-in-head angular velocity, with a provisional 60-degree-per-second threshold, a duration of 20–100 ms, at least two sampled intervals, and observed

lower-velocity samples on either side. Intervals with head speed above 20 degrees per second or unavailable head pose are excluded. The reported rate divides event count by retained low-head-speed duration. The rate therefore describes saccades within this restricted window rather than across the whole search. These restrictions reduce head-turn ambiguity; event identity and sensor-level eye–head synchronization remain unvalidated.

The Index of Pupillary Activity (IPA) calculation follows the wavelet-based approach of Duchowski et al. [14]. Each eye is processed separately. Accepted pupil segments contain valid finite diameters, last at least two seconds, meet a median query-rate screen of 60 Hz, and pass an interval-regularity screen. Within each accepted segment, values are resampled at its median query interval; no interpolation crosses rejected gaps or invalid observations. Level-two sym16 wavelet detail coefficients yield thresholded modulus maxima, which are counted per retained second. The implementation records its thresholds, resampling, numerical tolerance, and nearest-neighbor boundary convention. When no segment qualifies, IPA is recorded as unavailable rather than zero.

Software tests verify event boundaries, query deduplication, invalid-data handling, head-turn suppression, and wavelet calculations; they establish computational correctness, not physiological validity. Fixation counts and saccade rates will enter the participant analysis only after independent event annotation, calibration checks, timing assessment, and held-out threshold evaluation. Pupil-based interpretation additionally requires checks for luminance, gaze angle, blink artifacts, segment boundaries, and task-related confounds, so IPA is an exploratory measure reported alongside NASA-TLX rather than a validated workload estimate for this apparatus. Final fixation duration on the target remains unavailable until validated eye events can be joined to target areas of interest; the last ray contact does not measure it.

3.4.7 Gaze Contact, Vicinity, Pointing, and Selection

Four event families are central to interpretation: a valid eye ray can intersect the target collider; a gaze direction can satisfy the larger target-vicinity rule; the right-controller ray can intersect the target; and a trigger press can select the target. These events can occur at different times and need not occur in that order in every recorded sample sequence.

The first gaze-contact latency uses the first valid recorded target-ray intersection after reveal. The last gaze-contact time uses the last such intersection before trial end. Neither is a fixation duration. If no contact is observed because of a blink, acquisition gap, or a path that only enters the broader vicinity, the contact field remains missing.

Vicinity qualification is an interaction rule rather than a fixation classifier. In the current system, a target can be acknowledged after 100 ms of valid qualifying evidence, with overlapping regions resolved to a single closest candidate. The rule trades exact collider intersection for a tolerance intended to make feedback usable, but an overly large region can produce premature acknowledgment even when the timing gate behaves correctly. The tolerance value will therefore be logged and set through piloting; the default ring in the synthetic viewer screenshot is illustrative rather than a validated study setting.

Controller pointing is a distinct input observation, but its spoken acknowledgment also depends on gaze: it requires a qualified one-second controller hold with valid gaze in the target vicinity, and it plays when the speech scheduler permits. Before the hold completes, a valid controller hit on the target already suppresses contradictory directions pointing away from it, such as “go back down”, without announcing that the hold has qualified. Only a correct trigger selection completes the search successfully; stops and timeouts end the trial with other outcomes.

Figure 3.11 contrasts the two acknowledgment events with trial completion.

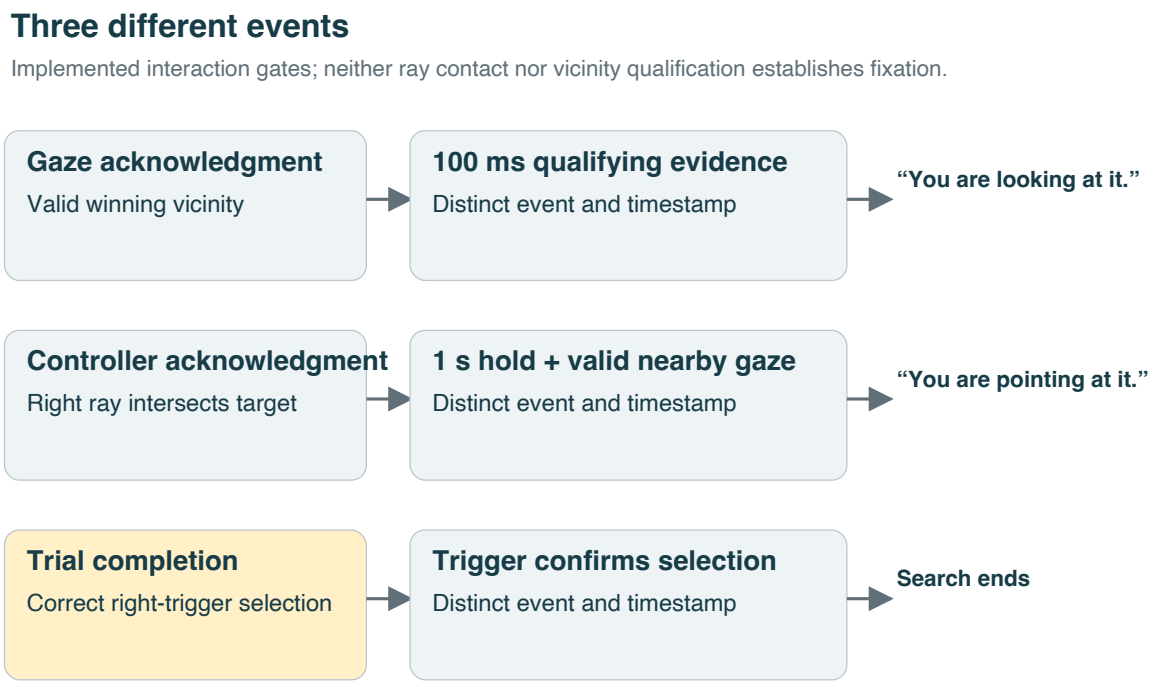


Figure 3.11: Implemented interaction distinctions. Gaze acknowledgment uses a 100-ms qualifying vicinity gate; pointing acknowledgment uses a one-second controller hold with required valid nearby gaze. A correct trigger selection completes the trial independently of whether either acknowledgment was spoken. These are interaction gates, not validated fixation measures.

3.4.8 Acquisition Rate, Validity, and Gaps

A render loop can query an eye-tracking API frequently without receiving a new hardware observation each time, so the effective acquisition rate is estimated from distinct source observations. If n unique observations span source times t_1 through t_n, a basic rate summary is $(n-1)/(t_n - t_1)$. Because a mean alone can conceal bursty delivery, the distribution of successive intervals, repeated timestamps, invalid observations, and long gaps accompanies that average.

Application timestamps describe when data were handled, whereas source timestamps describe when the tracking observation was produced, if the API exposes them reliably. Age or latency

across these clocks is therefore computed only once a clock-domain mapping is established. Where unique source timing is unavailable, that limitation is reported, and 120-Hz binocular delivery is not inferred from polling or headset specifications.

Spatial validation compares gaze direction with known independent targets. Angular error is computed from normalized gaze and expected target directions, with invalid observations excluded from the error distribution but retained in availability summaries. Precision and bias are assessed separately, since a stable offset and a noisy signal can produce different guidance failures even when their average angular error is similar. The validation procedure also documents target exposure, head pose, corrective lenses, and headset adjustments.

3.4.9 Motion, Coverage, and Instructions

Gross translation sums Euclidean differences between adjacent valid head positions, and gross rotation sums shortest quaternion angular differences; invalid intervals, excessive gaps, and pauses break the accumulation. Gross movement is a total, not a rate. If a separate movement-rate outcome is derived, its denominator is observed covered time rather than the entire nominal session, so that runs with different data loss remain comparable.

Fixed-axis components derived from rotation vectors differ from Euler yaw, pitch, and roll and are labeled by their actual calculation. Head motion is also only an indirect measure of swivel-chair rotation, because the head can move relative to the torso and chair.

Instruction counts are based on actual software playback starts; requested cues, rejected model answers, and prefetched clips are not counted as delivered instructions. Opening cues, search guidance, acknowledgments, reminders, and success speech have separate categories. The planned total of delivered active-search assistance includes directional guidance, gaze/pointing acknowl-

edgments, and selection reminders, with each category also reported separately, while opening and success speech remain outside that total. Interrupted speech counts as started speech but retains its completion status, since the participant may not have heard the full utterance.

3.4.10 Sequence-Level Latency

For a qualified event, let t_q denote the earliest supported qualification time and t_p the actual software playback start of the corresponding acknowledgment. The primary software delivery delay is

$$L_{\text{software}} = t_p - t_q. \tag{3.6}$$

This quantity excludes acoustic latency at the participant's ear, since audio-device buffering and the physical sound path require separate measurement. Gaze and controller acknowledgments are each measured from their own qualification time, because measuring a controller acknowledgment from first gaze contact would inflate or misattribute its delay.

Repeated requests about one continuously qualified event form a single episode, so a sequence can include several decisions by Jev to wait without creating several independent missed confirmations. An episode ends with playback, disqualification, trial termination, or a defined reset, and the evaluation reports how those endpoints were assigned, marking incomplete evidence as unknown.

Where logs support attribution, waiting is divided into model response, explicit model deferral, scheduling, existing speech, and audio readiness. Because these intervals may overlap, they are not summed unless they are represented as a nonoverlapping partition of the critical path. The full event timeline remains the authority when aggregate attribution is ambiguous.

Figure 3.12 separates intentional evidence qualification from subsequent delivery delay.

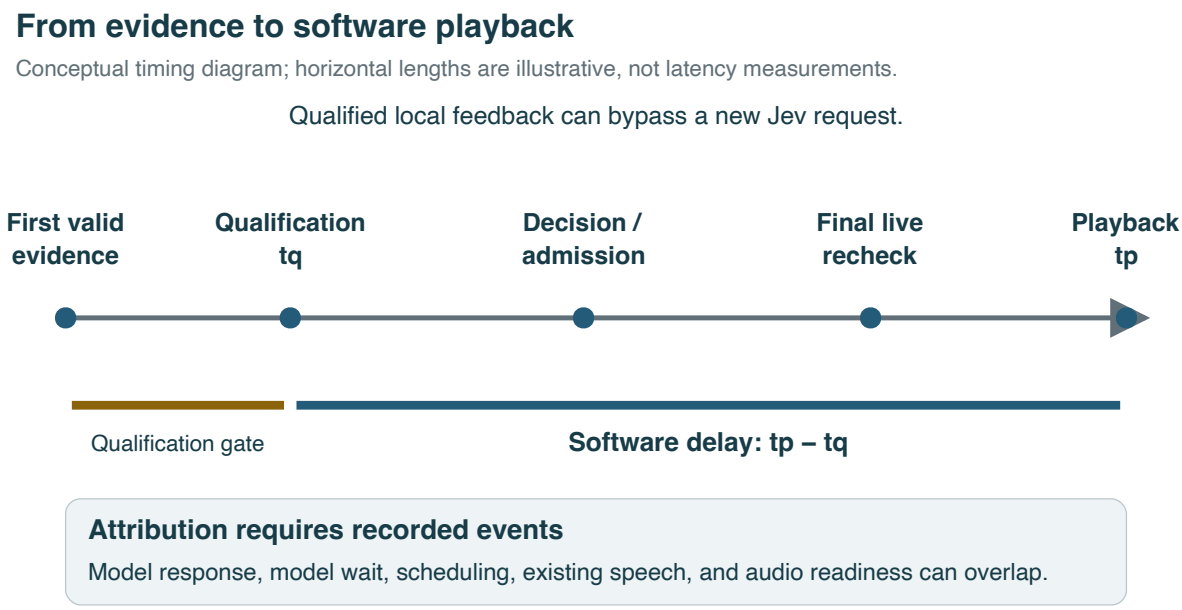


Figure 3.12: Conceptual event timeline, not a measured trace. Software delivery delay begins at supported qualification and ends at actual software playback start. The preceding qualification gate and the unmeasured acoustic path are separate quantities. Qualified local feedback can avoid a new cloud request.

Table 3.3 lists the principal timing variables with their endpoints and interpretive limits.

3.4.11 Questionnaire Joins and Versioning

Final questionnaire administration is separate from the block to which a response refers. A response collected after both blocks retains both its final-administration timing and a Block 1 or Block 2 reference, and the voice identifier is joined through the saved condition order rather than inferred from block number. Practice is excluded from measured block outcomes even though it uses the first voice.

Table 3.3: Timing variables for planned analysis and system evaluation. Cross-clock comparisons require an established clock mapping; unavailable durations are left missing rather than set to zero.

Quantity	Endpoints	Interpretation limit
Active completion	Reveal to correct trigger, less union of pauses	Excludes readiness and opening cue; unfinished trials remain unfinished.
First gaze contact	Reveal to first valid target-ray hit on active clock	Intersection, not fixation or vicinity acknowledgment.
First pointing contact	Reveal to first tracked right-ray hit on active clock	Not the one-second hold or trigger selection.
Jev response latency	Request dispatch to response receipt	A response can be deferred, canceled, or stale.
Confirmation delivery	Qualified evidence to matching actual software playback	Separate gaze and pointing episodes; not acoustic latency.
Live transfer lag	Source event to viewer receipt, if clocks are mapped	Describes transport, not headset decision delay.

Convenience exports repeat each block-level workload score on all 40 trial rows; the analytical dataset treats it as one participant-by-block observation and collapses duplicate joins when estimating questionnaire effects. Raw responses, score direction, scale bounds, item identifiers, instrument version, and missingness remain available.

The NASA-TLX adaptation uses 0–10 ratings, whereas older development exports used 0–100, so automated scoring relies on explicit range metadata. A separately named normalized score supports comparison while raw values are kept unchanged, and legacy responses with unknown Performance anchors are left unscored rather than assigned a direction. Complete-subscale scoring for the IMI and agent-perception measures will be verified before confirmatory analysis.

3.5 Measures and Scoring

3.5.1 Task Performance

Active completion time.

The primary outcome is elapsed active-search time in seconds from object reveal to the correct right-trigger selection, excluding the union of recorded pause intervals overlapping that period. Section 3.4 specifies the clock equation and handling of overlapping pauses. Lower values indicate faster completion. Readiness, opening guidance, post-selection feedback, questionnaires and between-trial breaks do not contribute to this outcome. An unfinished trial retains its observed exposure and outcome but has no successful completion time. The full interval from opening-cue onset to selection is also defined to describe the complete instructed task duration.

Target contact and selection errors.

Time to first target gaze is the active-clock time of the first valid, recorded gaze-ray intersection with the target, and time to first controller contact uses the first tracked right-controller ray–target intersection. Last target-gaze contact time denotes the last observed valid target intersection during the trial, not a fixation duration or a spoken confirmation. These event times remain distinct from trigger selection and from the larger gaze-vicinity region used by the guidance system. Incorrect selections are counted as trigger presses that select non-target objects. If no valid target contact is observed, its latency is missing rather than zero; all contact times retain their source and sampling resolution.

3.5.2 Starting Geometry and Search Behavior

Starting angle, height and distance.

The pre-cue snapshot saves a reference origin (the eye origin at readiness completion), readiness-cross position and target position. Starting theta (θ_0) is the signed angle about world up between the horizontal origin-to-cross and origin-to-target vectors; its magnitude $|\theta_0|$ quantifies horizontal turn demand, and its sign preserves direction. Degenerate horizontal vectors are marked unavailable. The primary analysis adjusts for $|\theta_0|$ rather than dividing completion time by degrees. Nominal wall-angle assignment, exact pre-cue theta and post-reveal head-to-target error are separate variables. Recordings made before the snapshot was added carry only nominal theta and therefore lack θ_0.

Signed target-height difference is $\Delta h = y_{\text{target}} - y_{\text{cross}}$, in metres; positive values indicate above-cross targets. Euclidean target distance is the three-dimensional distance between target center and the first valid tracked head position after reveal, with that observation's timestamp retained. All positions used in a calculation share a recorded coordinate frame.

Head movement and zone coverage.

Gross head translation sums distances between adjacent valid head positions, in metres, and gross head rotation sums shortest quaternion angular displacements between adjacent valid orientations, in degrees. Fixed-axis component totals are retained separately and labeled by their calculation rather than as Euler yaw, pitch and roll. Invalid intervals, pauses and excessive sample gaps break accumulation; the analysis does not interpolate across them. Covered time and gap counts accompany totals. These measures describe head movement, not independently tracked chair rotation.

Distinct walls visited, wall transitions and revisits summarize search coverage using stable valid gaze observations and a versioned zone-assignment rule. Transitions across tracking gaps are not inferred.

Instructions and guidance timing.

The delivered active-search assistance total counts playback starts for directional guidance, gaze/pointing acknowledgments, and selection reminders, including delivered repeats; these categories are also reported separately, as specified in Section 3.4. Requested repeats, delivered repeats and interrupted utterances are retained separately; model requests, prefetched audio and success speech are excluded from delivered search instructions. Opening cues and success feedback retain separate event labels. Timing analysis distinguishes evidence qualification, decision-request latency, model deferral, scheduling delay and audio onset. Gaze confirmations and controller confirmations are evaluated separately against their corresponding gates; the controller gate includes both its hold and the required nearby valid gaze.

3.5.3 Subjective Measures

Workload.

Six NASA-TLX dimensions—mental, physical and temporal demand, performance, effort and frustration—are rated separately for each voice block during the final survey, after both blocks. The study uses a documented 0–10 adaptation without pairwise weighting. Demand, effort and frustration run from 0 = Very low to 10 = Very high. The Performance prompt asks how successful the participant was and runs from 0 = Failure to 10 = Perfect. This favorable-direction response is a deliberate local adaptation of the opposite orientation in the NASA rating form. All six raw

responses are retained; higher raw Performance values indicate better perceived performance. Before calculating the workload mean, only Performance is reverse-scored once as $10 - r_{\text{Performance}}$. The unweighted mean uses the other five raw ratings and this reversed Performance value, and is calculated only when all six responses are present. Higher mean scores therefore indicate greater workload or poorer perceived performance. The source instrument is Hart and Staveland's NASA-TLX.

Motivation and activity experience.

The Intrinsic Motivation Inventory contributes interest/enjoyment (7 items), effort/importance (5), pressure/tension (5), and value/usefulness (7). Responses range from 1 = Not at all true to 7 = Very true. Published reverse keys are applied before averaging each complete subscale. Interest/enjoyment represents intrinsic motivation; the four subscales are not combined into an overall IMI score. Activity wording refers to the completed search block, with task-specific adaptations documented.

Agent perception.

Twenty-one items from Guo et al. (2024), Appendix A, Table A1 assess co-presence (Q1–6), perceived intelligence (Q13–18), appealing/eerie/familiar impressions (Q20–22), and perceived anthropomorphism (Q37–42). "Virtual character" is adapted to "voice agent." Responses range from 1 = Very strongly disagree to 7 = Very strongly agree. The three six-item constructs have separate complete-scale means. Q20–22 are analyzed individually, without a combined eeriness score.

Voice experience and reliance.

Twelve custom 1–7 agreement items assess voice similarity, responsiveness, helpfulness, distraction, familiarity, trust, comfort, uncanniness, reliance, naturalness, willingness to follow and motivation to act without guidance. Voice similarity serves as a manipulation check. Voice liking and retrospective percentage of guidance use each use 0–100 responses with separate unable-to-answer options. These 14 numeric items are analyzed individually, as they do not form a validated scale. Liking during voice audition remains separate from condition-specific liking collected at the end.

Debrief.

Open debrief questions explore perceived performance, guidance experience, own-voice comfort, possible use cases and voice preference; broad experience questions precede specific eeriness or inner-dialogue probes.

Each condition-specific set in the final survey contains 65 numeric responses (130 across both sets): six workload ratings, 24 IMI items, 21 Guo items and 14 custom ratings. Technical checks and optional text are additional. Questionnaire observations have participant-by-block scope even when repeated across the 40 trial summaries of a block, and missing answers remain missing rather than zero.

Table 3.4 summarizes the condition-specific battery. Counts refer to numeric items in one block-referenced set, not independent scales. Exact wording and reverse-key positions are preserved in Appendices A and B.

Table 3.4: Proposed final questionnaire battery for each referenced voice block. These ratings are administered after both blocks; setup voice ratings are separate.

Measure	Items	Scoring and interpretation
NASA-TLX adaptation	6	Six 0–10 dimensions; complete unweighted mean after reversing only raw Performance as $10 - r$. Preserve raw responses and explicit bounds.
IMI	24	Separate interest/enjoyment (7), effort/importance (5), pressure/tension (5), and value/usefulness (7) means. No overall total.
Agent perception	21	Three six-item means: co-presence, intelligence, anthropomorphism. Appealing, eerie, and familiar remain three single items.
Custom voice and assistance	14	Individual ratings of voice and guidance experience. No unvalidated combined score.
Total per block reference	65	130 numeric ratings across two references, before technical checks and optional text.

3.5.4 Measurement Quality

Tracking validity, observation age, data coverage, gaps and spatial-validation error accompany gaze-derived measures. Ray intersections and vicinity hits are reported as interaction events rather than fixations, and native eye-tracker sampling frequency is not inferred from render-frame observations. Pupil-based workload inference and fixation/saccade outcomes will be included only after independent validation. Trial performance and subjective summaries remain linked to separate raw gaze, motion and event records through participant, session, block and trial identifiers. Scale metadata distinguishes the present 0–10 workload form from any earlier exports using another range.

3.6 Planned Statistical Analysis

The analysis asks whether the two voice configurations differ in efficiency and experience. Active completion time is the primary outcome, workload and motivation are secondary, and agent expe-

rience, reliance, and debrief responses are exploratory. Because each participant experiences both voices, these outcomes are compared within participants, while guidance diagnostics describe exposure and technical context. Jev and gaze contingency are shared by both conditions, so the statistical comparison concerns voice configuration rather than the independent value of those components.

3.6.1 Primary Contrast

The primary estimand compares active completion time under the self-similar and neutral voices, standardized to the same starting-difficulty distribution. A trial-level mixed-effects analysis of log completion time accounts for participant clustering, voice condition, period/order, trial progression and absolute pre-cue starting theta. The analysis reports the voice time ratio and uncertainty using a two-sided contrast.

Nonlinearity in starting difficulty is represented by a prespecified low-complexity function of theta, while nominal-angle strata support balance plots and a categorical sensitivity analysis. Zero-degree trials remain in the analysis, and completion time is not divided by angle. Before confirmatory recruitment, simulation is used to finalize the model's random-effects structure, angle function and smallest meaningful effect. The participant count will be justified under that same model rather than by treating repeated trials as independent participants.

Figure 3.13 illustrates the planned units of observation. Trial-level measurements are nested within participants, while retrospective ratings remain participant-by-block observations.

Table 3.5 summarizes the proposed outcome hierarchy and units of analysis. Final contrasts, missing-data handling, and multiplicity families will be specified in the preregistration.

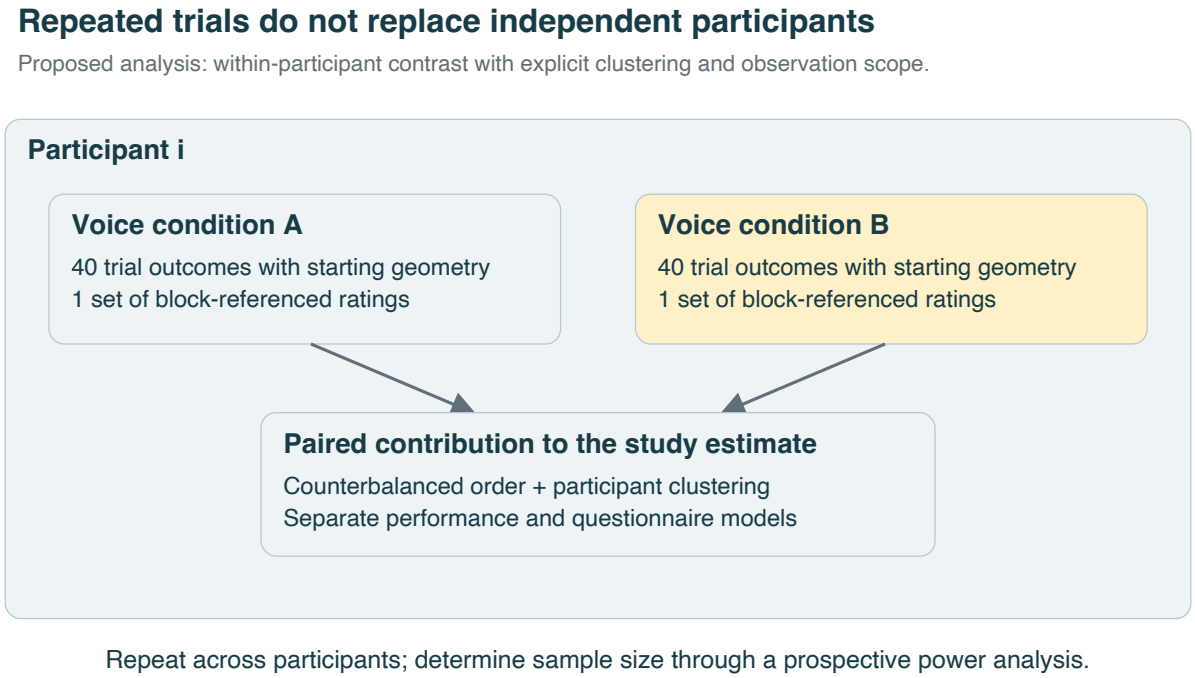


Figure 3.13: Proposed analysis scope. Each participant contributes repeated trial outcomes and two block-referenced questionnaire sets. The analysis accounts for counterbalanced order and participant clustering, and sample size is justified at the participant level rather than by trial count.

Table 3.5: Outcome hierarchy and observation scope. Guidance-delivery metrics characterize the system and are reported separately from participant outcomes.

Role	Unit	Planned interpretation
Primary performance	Trial within participant	Voice contrast in active time, adjusted for starting geometry and design factors.
Secondary experience	Participant by referenced block	Workload and separate IMI subscales, retaining paired scope.
Exploratory perception	Participant by referenced block	Agent constructs, individual custom items, and voice-similarity manipulation check.
Safety	Participant over full session	Recorded discomfort reports, pauses, stops and incomplete exposure.
Engineering quality	Qualified event episode / request	Delivery delays, errors, repetition, and provider reliability with distinct denominators.
Qualitative experience	Participant response / coded excerpt	Debrief themes with a documented coding trail; descriptive rather than causal.

3.6.2 A Candidate Primary Model

The following model illustrates the intended estimand; the final specification will be fixed in the preregistration. For participant i, period b, and trial j, let $Y_{ibj} = \log T_{ibj}$ for successfully completed active-search intervals. A candidate specification is

$$Y_{ibj} = \beta_0 + \beta_V V_{ib} + \beta_B B_b + \beta_O O_i \tag{3.7}$$

$$+ f(|\theta_{0,ibj}|) + g(j) + u_{0i} + u_{Vi} V_{ib} + \epsilon_{ibj}. \tag{3.8}$$

Here V indicates self-similar rather than neutral voice, B identifies the second period, and O identifies assigned voice order. The functions f and g represent prespecified low-complexity effects of exact starting angle and trial progression. The index j denotes position within a block (1–40), and the period term captures the between-block change. The exact pre-cue angle comes from the snapshot described in Section 3.4; earlier development recordings provide only nominal wall angles. Participant intercepts and, if supported by the design and model stability, participant voice slopes represent clustering and variation in within-participant contrasts. With only one block per voice, the slope variance can also absorb block-specific session effects, so it does not isolate stable individual differences in voice responsiveness. The final random-effects structure will be selected on the basis of the experimental design and simulation rather than statistical significance [4].

Under this model, the exponentiated voice coefficient, $\exp(\beta_V)$, is a conditional multiplicative time contrast, which need not equal a marginal arithmetic mean ratio. The primary estimand is therefore the marginal contrast standardized over the prespecified common starting-difficulty distribution, and the conditional coefficient is reported separately as a model parameter. Predictions on the original seconds scale require an explicit retransformation procedure and uncertainty estimate.

A two-period design supports adjustment for period and sequence, but cannot identify every possible asymmetric carryover mechanism independently. For example, using a voice during practice could affect familiarity in its first block, and the second block could benefit from task learning. Period and order terms therefore address these influences only partially, and the report describes residual carryover as a limitation.

3.6.3 Sample Size and Precision Planning

Forty trials per block increase the information available within each participant, but a participant's 80 trials are correlated observations, so precision depends on the number of participants as well as the number of trials. The power analysis will reflect the planned trial-level model, between-participant heterogeneity, within-participant correlation, geometry distribution, incomplete trials, and the effect size worth detecting.

The planned simulation generates participants with both voice blocks and the intended nominal-angle quotas, varying plausible residual variation, participant intercept variation, voice-slope variation, and missingness. Each simulated dataset is fitted with the same model intended for the final analysis, and convergence, bias, interval coverage, and power or precision are summarized across candidate enrollment sizes. The smallest meaningful time ratio and its supporting assumptions will be documented before recruitment. Synthetic pilot values used to test the workflow are labeled as synthetic rather than presented as empirical variance estimates.

Pilot data can inform variance assumptions only if a prespecified rule determines whether those observations enter the confirmatory sample. Each session records a software-version and protocol identifier, so pilot data collected under a different guidance policy can be distinguished from data collected with the final study version.

3.6.4 Secondary Outcomes and Multiplicity

Condition-specific ratings are collected retrospectively after both blocks, with survey timestamps retained, and each rating set refers to one participant-by-block observation. Because ratings are not collected immediately after each block, recall and recency effects are considered when interpreting condition differences. Individual workload dimensions and complete unweighted workload means are summarized by participant and block. IMI subscales and the three six-item agent-perception constructs remain separate scores, and the additional Guo Q20–22 appealing, eerie and familiar ratings are analyzed individually rather than combined into an eeriness score. Voice similarity serves as a manipulation check, while custom experience ratings and self-reported guidance use are also analyzed individually.

The analysis retains paired observations and accounts for period where appropriate. The choice between ordinal-item and approximately continuous scale models will be justified for each measure before confirmatory analysis.

Multiplicity families, secondary contrasts, and adjustment rules will be specified before confirmatory analysis. Holm's procedure can be applied to a prespecified family of confirmatory secondary comparisons [21] once the membership of that family has been fixed. Exploratory agent-perception and custom-item results are labeled as exploratory and reported with effect estimates and uncertainty rather than as a list of significance claims.

A participant who does not recognize the clone provides information about the strength of the manipulation and is not excluded solely on that basis; similarly, low liking of either voice is an outcome rather than grounds for exclusion. Sensitivity analyses may examine synthesis quality or recognition, but conditioning on experience that follows assignment changes the interpretation of the contrast, and that change is stated.

3.6.5 Data Screening and Missingness

Data screening first establishes structural integrity: unique participant/session/block/trial keys, consistent voice assignments, chronological reveal and selection events, and required geometry provenance. It then assesses observation coverage and protocol deviations.

A trial with sparse gaze coverage may have valid selection timing yet unreliable gaze-contact latency, so exclusions are measure-specific where justified. Completion time can remain valid when gaze contact is missing, and a complete motion stream cannot recover missing voice playback events. Each analysis reports its own denominator and reasons for missingness.

Stopping, timeout, withdrawal, and technical failure will be handled under prespecified rules. A complete-case time model estimates completion times among observed successful trials under its assumptions, and it may not describe all assigned trials if incomplete outcomes differ by voice. The final plan will therefore also define a complementary completion-probability or time-to-event analysis for use if incompletion is consequential. The protocol sets the active-search deadline at 25 seconds. Deadline-expired attempts count as unsuccessful outcomes for completion probability and are right-censored at 25 active seconds in a time-to-event analysis, rather than treated as 25-second completion times. Earlier development recordings predate this rule and retain their original exposure.

Raw attempts, incorrect selections, pause intervals, timeouts, technical failures and withdrawals are all retained, and stopped trials are not coded as successful completion times. The deadline outcome is recorded separately from technical failure and withdrawal. The remaining treatment of those interruptions and the missing-data estimand will be specified before collection; slow successful trials within the deadline are not removed solely for being slow. Missing responses are not replaced with zero; questionnaire scores follow the complete-subscale rule, with item-level missingness reported.

3.6.6 Behavior as Outcome Rather Than Automatic Adjustment

Instruction count, gaze path, and controller behavior can be consequences of voice condition. Including them automatically as primary-model covariates could remove part of the effect the study intends to estimate. The primary comparison therefore adjusts for pre-cue difficulty and design variables. Voice-by-theta variation, selected neutral voice, acoustic quality, guidance timing, and search behavior enter exploratory or sensitivity analyses with their assumptions reported.

For example, slower search can produce more opportunities for guidance, and more guidance can itself affect search time. A correlation between instruction count and duration does not establish that speech caused delay.

3.6.7 Guidance Delivery Evaluation

The delivery analysis distinguishes event qualification, decision latency, model deferral, scheduling delay and actual playback, and it evaluates gaze and controller confirmation separately. Sequence-level review includes what happened before and after each cue. Automated checks and ratings from an offline language-model evaluator (Appendix C) serve as advisory system-level evidence and are distinct from participant perception scores.

3.6.8 Additional Performance and Eye-Behavior Figures

Missed-target percentages complement completion-time distributions, with the number of completed, timed-out, technically interrupted, and unresolved attempts shown explicitly. For the participant study, paired participant summaries by voice condition are preferable to treating all trial rows

as independent observations. Development recordings collected without a deadline are reported separately from searches governed by the 25-second deadline.

After independent detector validation, possible additional figures include fixation-count distributions, saccade rates with their valid-time denominators, and IPA distributions with retained pupil coverage, each separated by voice condition and participant. Last-target-fixation duration requires validated events and a target-area join. No EEG or ERP plot can be generated from the current headset logs. Until then, the candidate eye-measure figures in Section 4.1.4 serve only to inspect the processing pipeline and support no inference about voice effects or workload. Threshold sensitivity and missing-data summaries accompany each derived outcome.

3.6.9 Qualitative Analysis and Mixed Evidence

The debrief begins with broad experience questions, followed by balanced probes about voice preference, perceived performance, comfort, and potential uses. The coding process records the initial codebook, revisions, coding decisions, and independent review of a subset. Contradictory or minority accounts are retained rather than forced into one favorable theme.

Reported reliance and observed behavior are considered together without treating either as a perfect measure of the other. The final report separates quotations, coder interpretation, and behavioral evidence.

3.6.10 Safety Outcomes and Manipulation Checks

Safety reporting describes recorded discomfort reports, pauses, stops and incomplete exposure, including the reason for each interruption when available. These observations document tolera-

bility and session completion rather than a standardized pre-to-post symptom score. The stopping procedure will be finalized before participant collection.

The similarity manipulation check reports the paired difference between self-similar and neutral-voice ratings with uncertainty. A group-level decision rule and any practically meaningful difference will be set in the preregistration. Because no equivalence test is specified, a nonsignificant voice contrast is not interpreted as equivalence.

CHAPTER 4: RESULTS

The results chapter returns to the question posed in the introduction: does guidance spoken in a synthetic version of the participant's own voice differ, in efficiency or experience, from guidance spoken in the neutral voice the participant selected? The chapter first establishes how the comparison is delivered, then addresses search efficiency, workload, motivation, and exploratory measures of voice experience. Because confirmatory participant data have not yet been collected, the sections below describe the reporting structure for each result and separate those pending results from the development evidence available for the apparatus.

4.1 Recorded Development Results

The descriptive results in this section come from development headset recordings rather than simulated trials. The analysis includes all nine locally mirrored `DEV_NEUTRAL` recordings dated October 19–20, 2026 (Runs 20–28), with 27 started searches and 21 recorded correct selections. These sessions used the neutral-male development mode; the eight available trial summaries identify the Eric voice and a 168-object layout. This earlier development configuration differs from the planned participant-study stimuli. Each recording comes from a different software build, and the recordings do not constitute a sample of independent participants. The results therefore describe system operation during development; they estimate neither a self-similar versus neutral voice effect nor differences between software versions.

The analysis retains unfinished searches and uses the complete recorded prefix of Run 21, whose final JSON record is truncated. No missing endpoint is reconstructed. Source file hashes, recording identifiers, derived CSV tables, and the plotting script accompany the thesis sources under

`development-results/`. Twenty of the 21 completed-search durations also match the separately saved trial summaries within their 0.01-second rounding precision; Run 21 has no corresponding summary in the inspected mirror. This comparison checks agreement between exports, not independent timing accuracy.

4.1.1 Search Completion During Development

Figure 4.1 separates completed-search durations from recording coverage. Panel (a) shows each completed search, with box plots for runs containing at least three completions; runs with fewer completions show individual observations only. Runs with no recorded completion remain labeled with zero observations. Panel (b) includes all 27 started searches and identifies the six unfinished searches. Unfinished searches are treated neither as completion times nor as task failures, because a recording may end while the user is testing, pausing, or leaving the application. The logarithmic duration axis retains the longest completed search while allowing shorter searches to remain distinguishable.

The figures use compact, labeled panels and distribution summaries, following the presentation of descriptive results in Chiossi et al. [9]. Boxes span the first and third quartiles, the central line marks the median, and whiskers extend to the most extreme observations within 1.5 interquartile ranges. Every observation remains visible as a point, including values beyond the whiskers. Horizontal offsets separate points for legibility without altering measured values, and colors distinguish adjacent runs rather than voice conditions. No confidence intervals or significance annotations are estimated from these recordings.

Among the 21 completed searches, the median recorded active-search duration is 19.56 s, the 95th percentile is 44.43 s, and the maximum is 275.92 s. Percentiles use linear interpolation between ordered observations. The long completed search is retained rather than removed as an outlier:

these were unconstrained development sessions, and the logs do not establish that every active second represents uninterrupted task-directed effort. The plotted counter follows the application's search-timer logic, which excludes some intervals, and therefore differs from elapsed recording time. The distribution shows that the application records completed selections and variable search durations; it is not an estimate of participant efficiency under a standardized protocol.

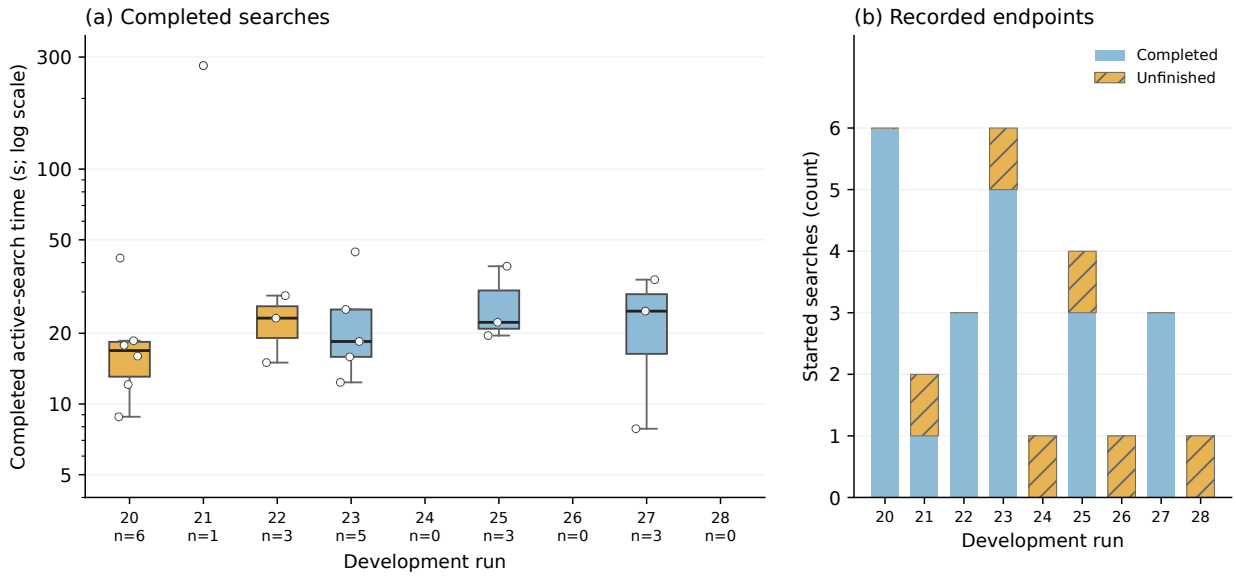


Figure 4.1: Recorded development searches. (a) Completed active-search times, with individual observations, sample counts, and quartile boxes where at least three observations are available; the vertical axis is logarithmic. (b) Counts of completed and unfinished searches. Unfinished searches are excluded from duration summaries, not assigned a completion time.

4.1.2 Speech Delivery and Earlier Waiting

Of 350 software playback-start events during the recorded search intervals, 339 have an unambiguous chronological match to a guidance request, audio request, and audio-ready event using the same recording, trial, and decision identifier. The remaining 11 playback events are excluded from the

latency distribution, rather than assigned zero delay, because the required join is missing or ambiguous. Requests that never produced playback are likewise outside this delivered-event distribution.

For the 339 matched events, the median interval from the delivered decision's request to software playback is 0.244 s, with a 95th percentile of 0.487 s and a maximum of 0.538 s (Figure 4.2). The median request-to-audio-request interval is 215.7 ms, compared with 17.9 ms from audio request to readiness and 1.0 ms from readiness to playback. The first interval can include decision-making and scheduling and so does not isolate Jev network latency. All timestamps mark software events rather than the onset of audible sound at the headset.

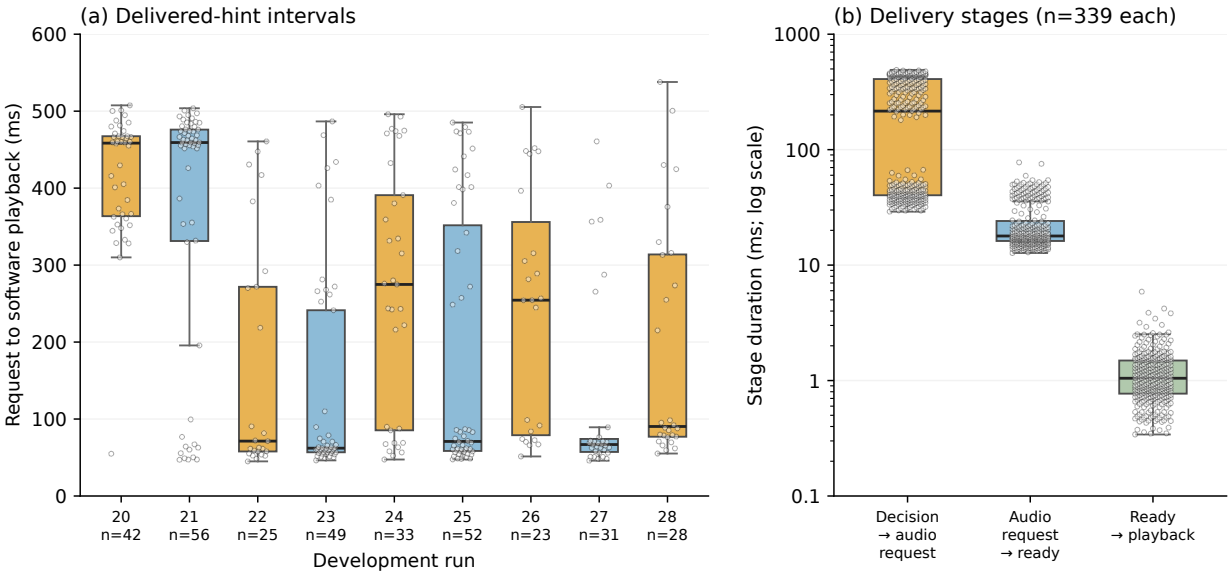


Figure 4.2: Software speech-delivery intervals for 339 matched playback events. (a) Request-to-playback distributions by run, with sample counts. (b) Delivery-stage distributions for the same events, using a logarithmic vertical axis. Points retain all observations; boxes and whiskers follow Figure 4.1. Earlier qualification and deferral, undelivered requests, and acoustic onset are outside this measure.

Short delivery intervals do not establish prompt feedback from the user's perspective. Figure 4.3 reconstructs the gaze-confirmation episode in Run 20, Trial 4. A gaze-qualified hint request is

recorded at session time 321.426 s. Three successive transmission snapshots record qualified target gaze, while controller-hold qualification is false. Jev returns *wait*, *wait*, and then *speak*; playback of "You've got it!" begins at 323.616 s. The interval from the first qualified request to playback is therefore 2.190 s, although the final request-to-playback interval is only about 0.352 s. The trace is one delayed episode with repeated requests rather than three independent confirmation failures. The snapshots document qualification at those instants but not uninterrupted fixation between them.

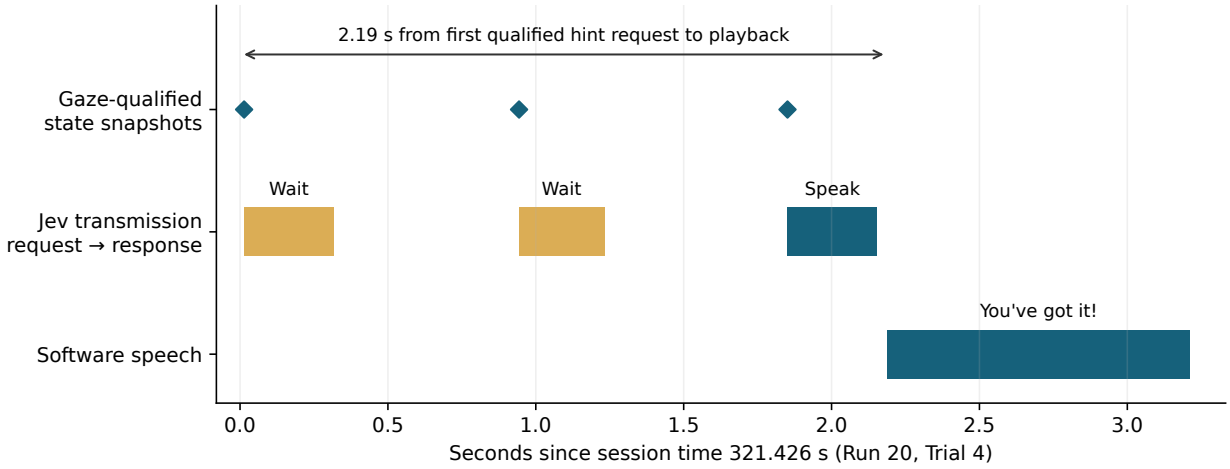


Figure 4.3: Recorded gaze-confirmation delay in Run 20, Trial 4. Diamonds mark qualified-state snapshots, bars mark transmission request/response intervals and software speech. Gaze confirmation is distinct from controller confirmation and correct trigger selection.

This case illustrates why sequence histories are needed alongside aggregate delivery statistics. Earlier waiting remains visible even when the request that ultimately produces speech is processed quickly. Whether later software changes eliminated this behavior can be determined only from a comparable recording with the current version. No acoustic-latency or participant-perception conclusion is drawn from this trace.

4.1.3 Recorded Gaze Usability

Figure 4.4 summarizes 58,041 rendered-frame replay samples recorded during active search. A sample is counted as usable when the saved gaze-device availability flag is true and the replay's gaze-usable flag is set. This is the application's recorded usability decision, which includes tracking and freshness checks, rather than a new physiological classification. Samples marked unusable are retained, and absent or indeterminate flags would be classified as unknown.

Across the nine runs, usable-row fractions range from 63.75% to 99.97%. All inspected active-search rows have a usable/unusable classification; however, intervals with no recorded rows remain unobserved and are not included in the denominator. The dominant logged invalidity reason is `both_native_eyes_invalid`. This diagnostic alone cannot distinguish among blinking, headset removal, poor fit, calibration, and software issues. Runs with long periods of unusable gaze warrant timeline inspection before they are used to assess guidance behavior.

The row fractions are diagnostic: they identify differences in the evidence available to the guidance system during these recordings, and they do not estimate spatial accuracy, native acquisition frequency, fixation quality, or the proportion of total search time with valid gaze. Together, the search, delivery, and usability plots characterize the apparatus ahead of the planned voice comparison; they do not address performance or experience differences between voices.

4.1.4 Exploratory Eye-Measurement Pipeline

A separate analysis uses the recorded eye-observation file for development Run 30 (October 23, 2026), which contains five recorded search identifiers in neutral-male mode. These searches predate the 25-second deadline, and their durations are not truncated. Figure 4.5 presents per-eye candidate

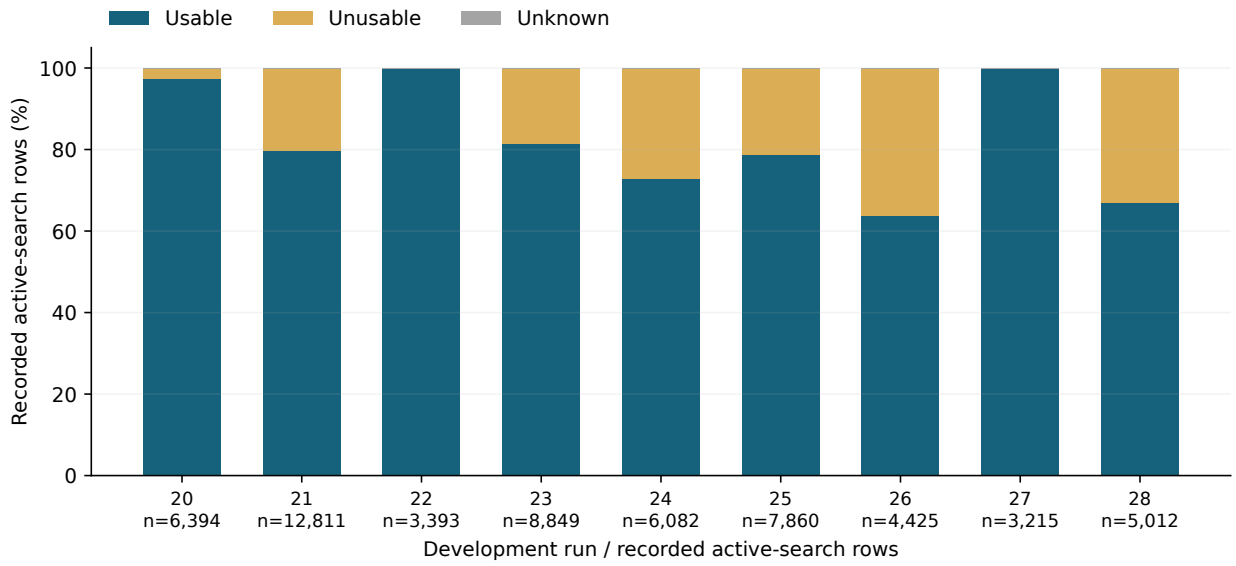


Figure 4.4: Recorded gaze usability during active search, with the number of replay rows shown for each run. These are rendered-frame observations, not independent native sensor samples. Unrecorded intervals are outside the denominator.

fixation counts, candidate saccade rates, exploratory IPA, and pupil duration retained for IPA. The figure uses only measured records; artificial test signals used to verify the code are excluded.

The strict pupil screens retain approximately 2.18, 2.25, 4.22, and 5.28 seconds per eye for the first four searches, with no accepted segment for the fifth. These short retained windows do not represent whole-trial workload, and unavailable IPA values are treated as missing rather than zero. Separate left- and right-eye values expose disagreement rather than averaging it away. The candidate fixation counts and restricted saccade rates likewise depend on detection thresholds and on the available query-time evidence. Three illustrative parameter variants are exported for sensitivity inspection, without choosing a setting on the basis of a favorable result.

The source hash, detector configuration, dependency versions, individual event intervals, summary CSV, and threshold-sensitivity table are retained with the thesis development artifacts. The compu-

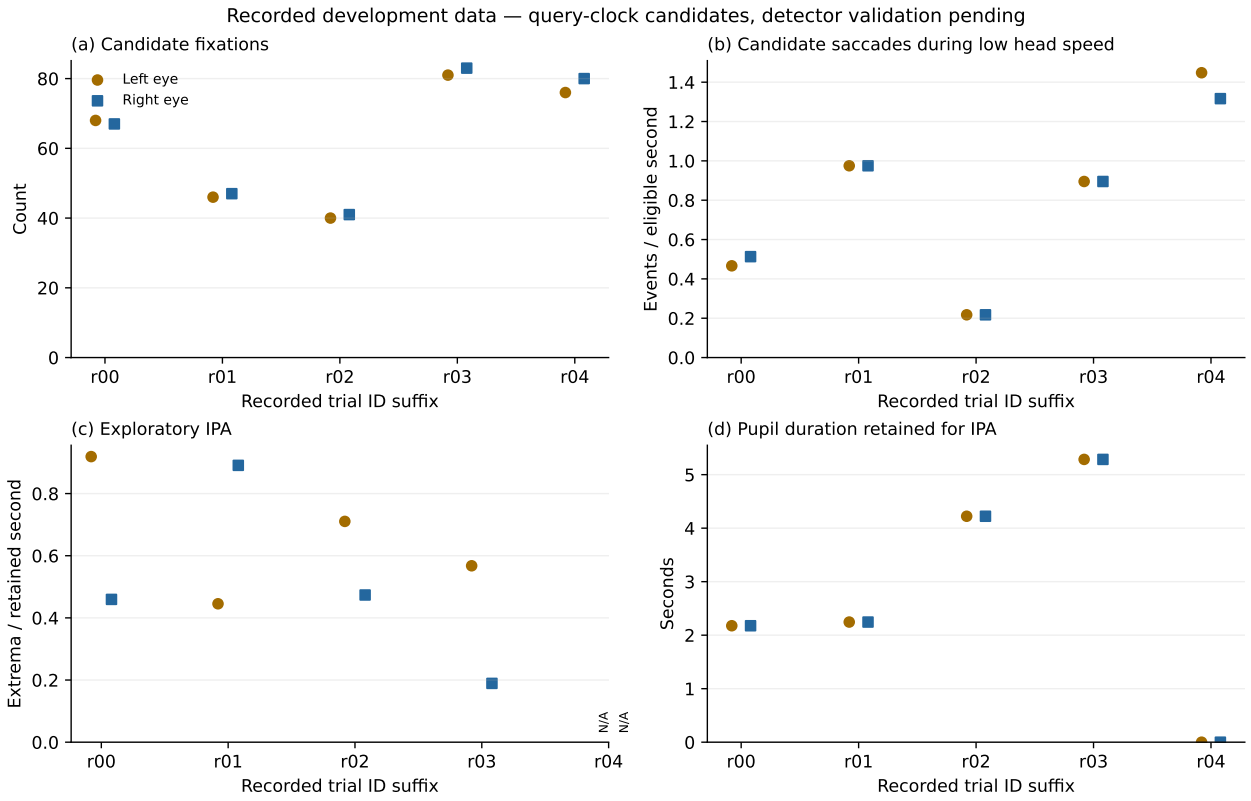


Figure 4.5: Run 30 development diagnostics. (a) Candidate fixation counts. (b) Candidate saccade events per retained low-head-speed second. (c) Exploratory IPA extrema per retained pupil second. (d) Retained pupil duration. Points represent separate eyes within recorded searches, not independent participants. The software-query clock does not establish 120-Hz sensor acquisition, and the classifications require independent validation.

tational checks pass, but without an independently labeled headset sequence, event precision, recall, and duration error cannot yet be established. These results demonstrate an executable measurement workflow and its data limitations rather than participant-study findings.

4.1.5 Outcome Panels by Voice Condition

Figure 4.6 adapts the four-panel presentation of accuracy, missed targets, search time, and pupil activity in Chiossi et al. [9] to the present study's voice conditions. Each horizontal axis distinguishes

Neutral voice from *Self-similar voice*; AR/VR and physical/virtual subdivisions are not factors in this comparison. The observations come from development Run 30, whose five started searches all use the neutral voice; the self-similar position therefore contains no observations. Scheduled searches that never started do not contribute observations.

First-selection accuracy is 80%: four of the five completed searches contain no incorrect trigger selection before the correct selection. This is a run-level proportion among completed searches, rather than a distribution across participants or a measure of success before the 25-second deadline. Active-search completion times use the recorded search counter and retain their original values in seconds. Run 30 has no recorded 25-second deadline, so its missed-target percentage is unavailable; searches lasting longer than 25 seconds are not retrospectively classified as misses.

For the condition-level display, exploratory IPA is the arithmetic mean of the available left- and right-eye candidate rates within each search. Four searches have values from both eyes; the fifth has no eligible pupil segment and remains missing. This display summary is not a validated workload score; the separate-eye diagnostics and retained durations appear in Figure 4.5. Points in the time and IPA panels are repeated trials within one development run rather than independent participants, and the figure supports no estimate of a voice-condition effect. The participant-study comparison will use observations from both voices and paired participant summaries with consistent scoring and eligibility rules.

4.1.6 Eye Measures and Workload in Later Development Runs

Figure 4.7 applies the planned eye-measure and workload panels to three later development recordings from October 25, 2026: neutral-voice run 32 and self-similar runs 1 and 2. Unlike Run 30, these recordings include self-similar playback. All three use the angle-controlled schedule in its shortened development mode, the 25-second search limit, right-trigger selection, and the same guidance-

policy version, and each session ended before completing a block. The 22 resolved searches comprise 14 neutral searches (12 completed, 2 timed out) and 8 self-similar searches (7 completed, 1 timed out). They are repeated development trials rather than independent participants or paired sessions.

Candidate fixations come from the dispersion-based detector described in Section 3.4, applied to each eye within valid, contiguous search intervals; each plotted value is the mean of the two eyes' counts for one search. Candidate saccade frequency divides each eye's candidate saccades by its eligible seconds, which exclude intervals of rapid head movement, and averages the two rates. Both measures rely on the HTC query-receipt clock rather than a verified sensor capture clock, and neither detector has yet been validated against independent annotations. Last target fixation duration is unavailable because it requires a validated join between eye events, the target's area of interest, and the selection time. Raw NASA-TLX is unavailable because these development sessions collected no questionnaire responses. Both panels are kept so that the missing measures remain visible.

Figure 4.8 reports, for each search, the share of active search time eligible for each detector, taking the lower value of the two eyes. Fixation-eligible coverage was at least 94% in every search except one neutral search at 47%, and saccade-eligible coverage ranged from 43% to 93%. Because fixation counts are totals over eligible time, they scale with search duration and coverage, and the self-similar searches were longer (median 19.7 s versus 13.4 s). Voice, session, search duration, and coverage therefore vary together, and the differences visible in panels (b) and (c) cannot be attributed to voice; no inferential test is applied. None of the three source recordings closed cleanly, and two end in a truncated record that was omitted. These figures show that the measurement pipeline runs end to end on headset recordings with both voices; they do not estimate a voice effect.

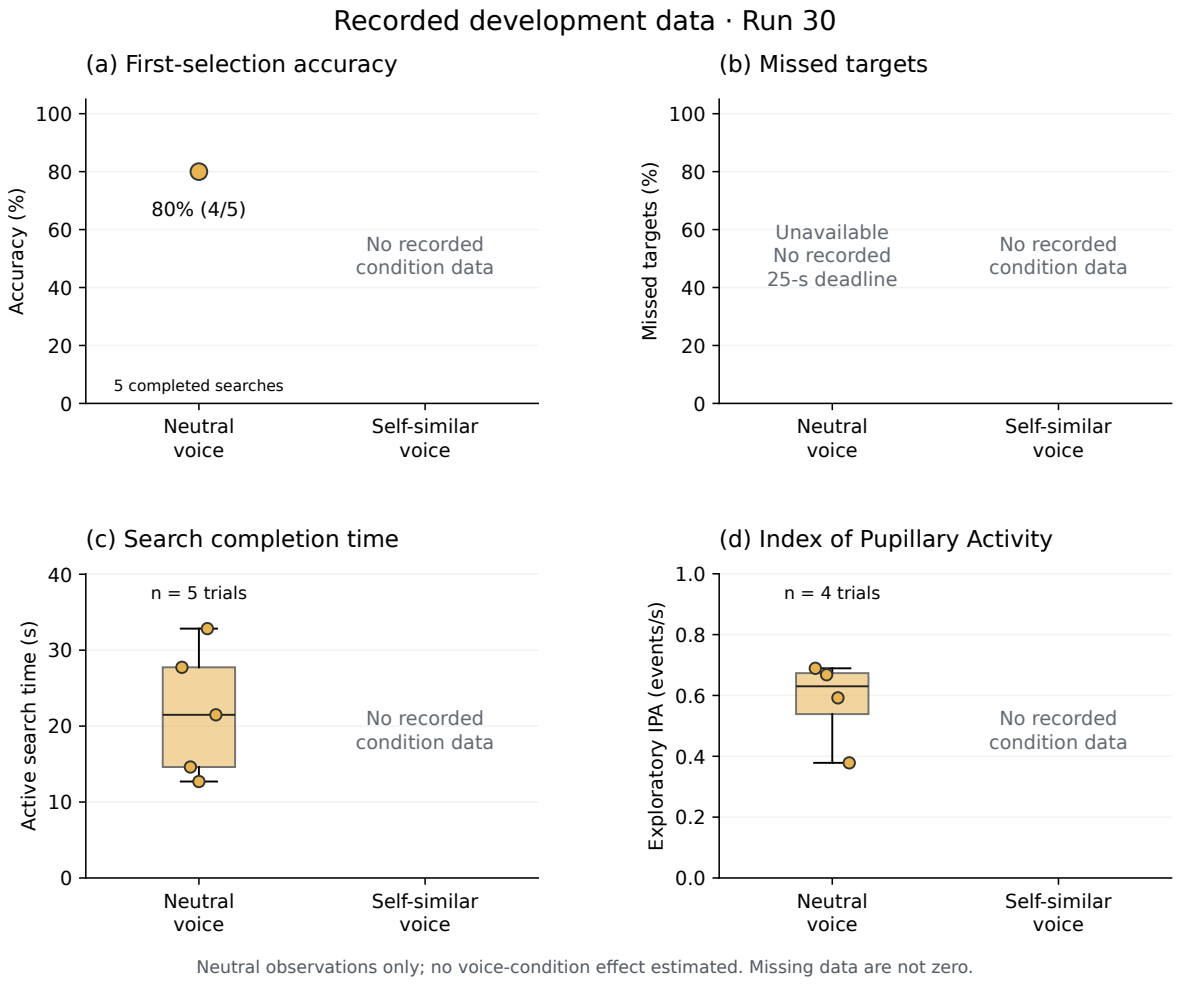


Figure 4.6: Development outcome panels arranged for the neutral versus self-similar voice comparison. (a) First-selection accuracy among completed searches, shown as one run-level point. (b) Missed targets, unavailable for this recording without a recorded deadline. (c) Active search completion time. (d) Exploratory IPA averaged across available eyes within each trial. Boxes in (c) and (d) show quartiles and median, with 1.5-IQR whiskers; all observed trial values are shown. Unavailable values are not plotted as zero. No self-similar observations or inferential comparisons are included.

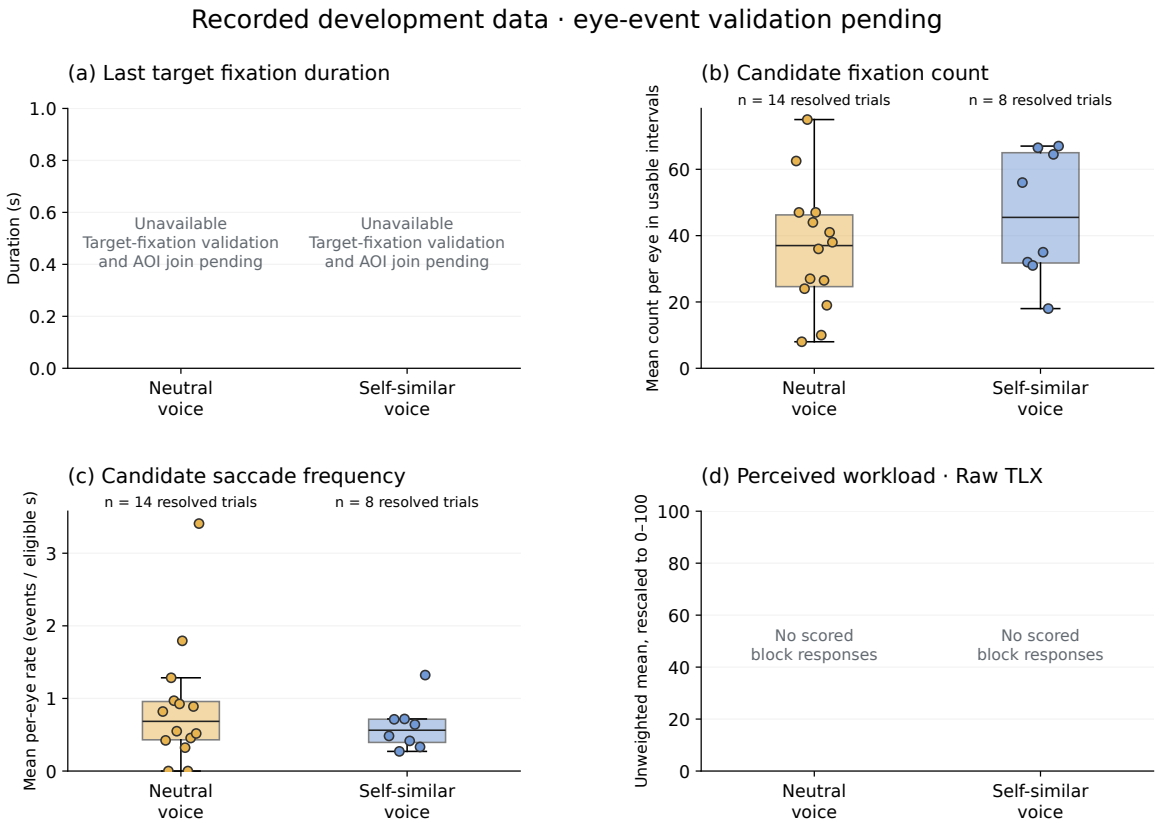


Figure 4.7: Eye-measure and workload panels from October 25 development recordings. (a) Last target fixation duration and (d) raw NASA-TLX are unavailable and are left empty rather than plotted as zero. (b) Candidate fixation count, mean of the two eyes over eligible intervals. (c) Candidate saccade frequency, mean of the two eyes' events per eligible second during low head speed. Each point is one resolved search (neutral run 32, $n = 14$; self-similar runs 1 and 2, $n = 8$); boxes show quartiles and median with 1.5-IQR whiskers. The detectors are not yet validated, and no inferential comparison is made.

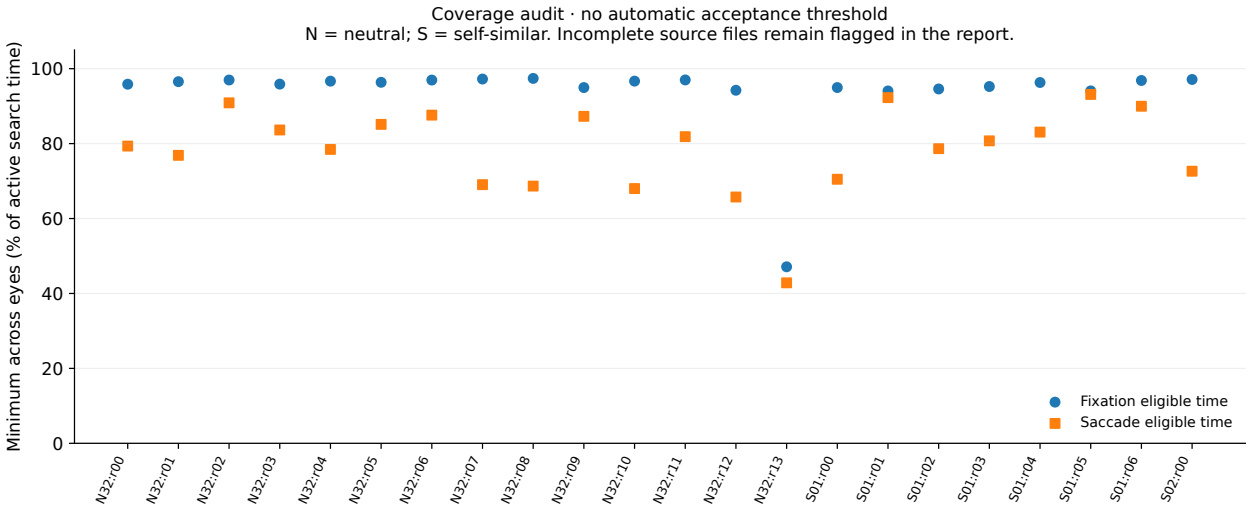


Figure 4.8: Eye-measure coverage for each search in Figure 4.7, as the percentage of active search time eligible for fixation and saccade detection, using the lower value of the two eyes. N32 denotes neutral run 32; S01 and S02 denote self-similar runs 1 and 2. No acceptance threshold is applied.

4.2 Planned Reporting of Participant Results

Participant reporting begins with participant flow, exclusions, incomplete sessions, and data coverage by condition and order. It then describes realized starting geometry and the similarity manipulation check before presenting the primary voice contrast.

The primary result includes the fitted time contrast, its uncertainty, the analyzed participant and trial counts, and model diagnostics. Secondary reporting covers workload and IMI subscales at participant-by-block level, followed by exploratory sections on agent perception, voice experience, guidance behavior, and debrief themes. Each table states its denominator and identifies missing observations.

Table 4.1: Planned result families and their status. No confirmatory participant outcomes are reported here.

Result family	Required evidence	Present status
Participant flow	Consented, started, completed, stopped; reasons	Not available
Manipulation check	Similarity by voice with paired scope	Not available
Primary performance	Adjusted time contrast and uncertainty	Not estimated
Workload and IMI	Versioned raw items and complete subscales	Not collected for this report
Agent experience	Individual and construct-specific ratings	Not collected for this report
Guidance quality	Sequence-level playback and evidence	Development traces reported; study comparison pending
Debrief themes	Recorded responses and coding audit	Not available

4.2.1 Results Organized by Research Question

Following the manipulation check, the reporting framework addresses each research question in turn. For the primary question, the estimand is the marginal time contrast standardized over a common starting-difficulty distribution, as defined in Section 3.6; here, *standardized* refers to that common distribution rather than to a standardized effect size. The estimate is accompanied by its uncertainty, the eligible participant and trial counts, and model diagnostics, following the analysis plan finalized before confirmatory recruitment. Secondary reporting covers the paired workload outcomes and the separate IMI subscales, using multiplicity families to be defined in the analysis plan. Exploratory reporting presents agent-perception and custom ratings alongside debrief themes and valid search measures. A contrast that cannot be estimated or proves inconclusive is reported as such rather than omitted.

Guidance delivery is reported separately through sequence coverage, qualification-to-playback delay, provider reliability, and categorized faults. Software tests and language-model ratings characterize the apparatus, not its effects on participants; conversely, a voice-condition difference does not show that every instruction was delivered correctly.

4.3 User Performance

Once participant data are collected, this section will report active search-completion time, incorrect trigger selections, and valid gaze/controller contact latencies by voice condition. Reporting includes paired participant summaries and the planned adjusted contrast, with sample counts and uncertainty.

4.4 User Experience

Once participant ratings are collected, this section will report condition-referenced workload, the separate motivation subscales, agent experience, voice liking and similarity, and debrief responses. No questionnaire responses or preference distributions are simulated.

4.5 Apparatus Verification

In addition to the development recordings in Section 4.1, software verification records document the apparatus. A verification run on October 20, 2026 passed schedule and voice-isolation checks confirming one practice trial in the first assigned voice, two measured blocks, and the nominal-angle quotas; it also passed 38 trial-export tests and seven workload scoring and plotting tests, and the Android application was built, installed, and launched successfully. These test counts come from that run and were not re-executed for this manuscript. The current software version, dated October 25, 2026, includes the pre-cue snapshot and balanced direction and height allocation. Its schedule checks pass across 50 participant identifiers and both voice orders, and its geometry checks pass across every starting wall and angle category.

4.6 Guidance Quality Supporting the Voice Contrast

Reporting describes gaze availability, actual speech exposure, delayed acknowledgments, interruptions, and unhelpful cues by voice condition and session. This account helps assess whether technical delivery problems affected one condition more than the other. The voice contrast is not automatically adjusted for behavior-dependent hint counts or waiting time: these quantities arise after

Table 4.2: Apparatus verification and outstanding validation.

Component	What was verified	Not yet established
Schedule and voice checks	One practice trial and nominal allocation of 40 trials in each of two blocks pass automated checks	Wearer completion of a full session
Trial export and TLX tests	Declared scales and retrospective joins are correct in tested cases	Validation of all questionnaires
Unity integration checks	Listed state and interaction cases pass	Robustness to all timing races
Android build/install/launch	Application built, installed, and started	Acceptance in a participant session
Exact pre-cue theta	Snapshot saved at readiness completion; geometry checks pass	Observation in a complete wearer session
Vertical allocation	20/20 and per-angle height quotas pass schedule checks	Target visibility and seated comfort
Voice-condition outcomes	None; no confirmatory dataset reported	Any performance or preference conclusion

voice assignment, may themselves depend on voice condition, and cannot be treated as baseline difficulty (Section 3.6).

Gaze acknowledgment, controller acknowledgment, and correct trigger selection remain separate events. Their sequence records document feedback delivery, whereas participant ratings and debriefing address how that feedback was experienced. Appendix C defines sequence tests and evaluation criteria for assessing the apparatus before interpreting differences between voices.

CHAPTER 5: DISCUSSION AND LIMITATIONS

5.1 What the Study Can Explain

Follow My Voice asks whether a synthetic voice resembling the listener changes search efficiency or experience relative to a neutral voice the listener has already chosen. Because participant results are pending, the interpretations below are conditional: they describe what the data could support, not which voice is better.

The discussion distinguishes three kinds of evidence. The primary question concerns measured efficiency, the secondary questions concern workload and motivation, and the exploratory questions examine participants' evaluations of the voices and their reported reliance on guidance. Agreement among these outcomes could strengthen a design interpretation, whereas disagreement could reveal a tradeoff that a single outcome would conceal.

5.2 Interpreting Efficiency and Experience Together

If completion time favors the self-similar voice, its interpretation depends on the estimate's uncertainty, the manipulation check, and guidance quality. A difference accompanied by systematic intelligibility or delivery problems in the neutral condition could reflect those problems rather than anything specific to self-similarity. If both conditions have comparable delivery quality and participants recognize the clone, the result would more directly support a voice-condition effect in this task, although resemblance would remain confounded with other differences between the chosen and synthesized voices.

Faster completion with the neutral voice would be consistent with a chosen external voice being easier to follow, but also with intelligibility or delivery problems in the cloned condition. Delivery records and acoustic-quality reporting could help distinguish these readings, although the study could not establish the mechanism without further evidence. Liking, naturalness, and reliance ratings may help characterize the experience in either case, but a correlation between such ratings and performance would not establish that they mediate the voice effect.

If completion times are similar, the width of the uncertainty interval becomes central. A wide interval would leave practically important differences unresolved, whereas a narrow interval near the null could constrain the size of a plausible effect under the tested conditions. A nonsignificant test does not establish equivalence. The analysis plan includes no equivalence test, which would require a meaningful bound and procedure specified before the result is examined.

A preference for self-similar speech may occur without a detectable improvement in completion time, while efficient search may coexist with discomfort.

Kao et al. examine an avatar voice in a setting where continued voluntary play can reflect engagement, whereas Guo et al. report favorable impressions alongside eeriness in virtual collaboration [25, 19]. Here, a voice instructs the listener through a fixed search task. Faster completion therefore addresses efficiency, while enjoyment, discomfort, and debrief responses address the experience of following it.

5.3 Practical Relevance and Transfer to Other Search Tasks

The applications that motivate this work indicate where guided search could be useful; this experiment does not validate guidance in those domains. Locating task-relevant items in maintenance or training, finding a book, and noticing relevant VR content share a need to connect assistance with

something in the environment [5, 54, 30]. The present study addresses a voice-choice question within that interaction and does not test industrial task completion, learning transfer, navigation safety, or narrative comprehension.

The practical value of the study depends on how its outcomes combine. Faster search accompanied by acceptable workload and a comfortable voice experience would support considering self-similar guidance for comparable tasks. A favorable experience without a detected speed difference could still motivate offering the clone as an optional voice, without supporting a claim that it makes search more efficient. Discomfort, intelligibility problems, or a preference for the neutral voice would instead mark limits to that option.

Transfer also depends on how much information users already have. Here, the participant must use spatial guidance to infer an initially unnamed target; in Zhang et al.'s book task, the target title was supplied [54]. A known-target task may allow people to continue independently and encounter the voice less often. Real work can additionally require recognizing meaningful parts, remembering procedural steps, moving through space, or coordinating with another person. Replication in those contexts would need to establish both the quality of guidance and the effect of voice choice.

5.4 Voice Identity and Acoustic Confounds

The voice conditions may differ in acoustic properties as well as perceived resemblance. The self-similar voice depends on recording quality and synthesis, whereas the neutral voice comes from a heterogeneous catalog, and the two may differ in pitch, prosody, speaking rate, pronunciation, and clip duration. Because identical wording and a shared provider do not guarantee matched acoustics, the study documents these properties and the intelligibility observed in pilot sessions at comfortable playback levels, without assuming exact equivalence.

The variety of chosen neutral voices is a deliberate consequence of the design: requiring every participant to hear the same neutral voice would replace the comparison with a preferred alternative by a comparison with a fixed one. The analysis retains chosen voice identifiers and describes their distribution, but does not fit voice-specific estimates that sparsely chosen voices cannot support. An exploratory association between a neutral voice and performance may reflect who selected it as well as its acoustic properties.

The self-similar manipulation applies no additional pitch shift, panning, or inner-thought treatment. Processing that makes cloned speech sound like an internal monologue is excluded from the study; applying such an effect in only one condition would change the manipulation and require a revised rationale and analysis.

5.5 Adaptive Guidance as a Shared Mechanism

Gaze-dependent sonification, staged spatial cues, and audiovisual assistance investigate different ways to connect a cue with a search target [33, 29, 5]. Follow My Voice adapts these precedents into spoken orientation and refinement so that the assigned voice participates throughout the task. Because the participant design compares only voices, a difference between conditions would not show that speech outperforms those techniques or that Jev improves search.

A shared policy does not produce identical spoken sequences. Because participants turn, pause, look, and point differently, the policy supplies different directions and can emit different numbers of hints.

Policy drift, however, would threaten the comparison: changes to phrases, timers, or provider behavior during data collection could alter assistance quality. The study therefore fixes the application build, configuration, phrase semantics, voice preparation settings, and decision-provider version

where possible, and any unavoidable service change will be documented and accounted for in the analysis.

Centralizing guidance evidence and transitions in a single local state machine makes guidance auditable but does not guarantee that it is correct. Calibration error can produce an incorrect observation, an unclear reference frame can produce an unhelpful phrase, and a stale response can pass an incomplete check.

5.6 Practice, Learning, and Order

One practice trial reduces setup burden but provides limited exposure to a complex interaction in which a participant must learn to hold gaze on the readiness cross, follow speech, use the chair comfortably, aim the right controller, and press the trigger to select. A single trial cannot ensure that all participants acquire these skills equally, and a practice-proficiency criterion will be established in pilot sessions.

Practice uses the first assigned voice, which aligns initial exposure with the first block but also couples practice familiarity with period. Counterbalancing helps distribute this coupling across voice conditions but does not prevent fatigue, learning, novelty, or asymmetric carryover, which are reported with period and order effects and any relevant sensitivity analysis.

5.7 Retrospective Questionnaires and Burden

The final-only survey avoids interrupting the two-block search sequence with a long questionnaire, but it also requires participants to recall Block 1 after completing Block 2. Ratings may therefore reflect recency, contrast, or imperfect recall. Each rating set identifies the block it refers to, and the

reminder linking that block to its voice will be piloted. Such reminders may aid recall but cannot guarantee its accuracy.

The condition-specific battery contains 130 numeric responses across both blocks, in addition to setup ratings, comparison questions, and debrief. Pilot sessions will therefore measure its completion time, missingness, and comprehension. Because participants may be tired or uncomfortable after exposure, researchers check whether they are comfortable continuing before administering the retrospective sets.

Adapting an embodied-agent measure to a voice agent can alter its meaning. Co-presence or anthropomorphism items may be understood differently without a visible character. The thesis documents source attribution and wording changes, reports the adaptation, and makes no claim of new psychometric validation from a small experimental sample. Custom reliance and naturalness items remain individual exploratory measures.

5.8 Gaze Quality and Feedback Coupling

Gaze serves both as an input to assistance and as a recorded behavioral signal. Tracking gaps can change the participant's experience and reduce measurement coverage, affecting both the outcome being measured and the evidence available to interpret it. Coverage and technical events are therefore interpreted alongside performance rather than treated automatically as pre-existing participant characteristics.

A highlighted vicinity candidate can help indicate what the system currently attends to, but it can also suggest a reference object that differs from the mathematical gaze ray. For example, an instruction to look lower can be consistent with a gaze ray above the target yet confusing if the target appears higher than a highlighted neighbor. Cues are therefore evaluated in their visible interaction

context, not solely against the sign of a geometric delta, and development-only rings and diagnostic markers will be excluded from the participant condition.

The proposed validity-aware stream and independent calibration checks address these risks, but headset-specific spatial performance and the rate of distinct gaze observations remain empirical questions. Validation results for the Vive Pro Eye cannot be assumed to hold for the Focus Vision, and an interaction threshold chosen for usability is not a validated measure of cognition.

5.9 Privacy, Consent, and Reproducibility

Coding participant identifiers does not make voice recordings anonymous, because both natural and synthesized speech can remain recognizable. Before participant use, the consent and data-management documents will be revised to describe recording, synthesis, cloud processing by ElevenLabs and Jev, storage, access, and deletion.

The thesis reports the relevant processing stages without including credentials or identifiable recordings. A reproducibility package can provide code, versioned schemas, example synthetic records, and de-identified summaries where authorized. Consent to participate does not extend to distributing cloned voices or raw participant audio, which would require a separate basis.

Cloud dependencies also affect replication. A future researcher may not obtain the same voice model or identical decision behavior. Preserving prepared-library metadata and actual playback records helps describe the original exposure, while a constrained phrase interface makes alternative providers testable, although provider changes must still be reported.

5.10 Future Work on Self-Similar Assistance

Future studies could examine repeated exposure, task context, and the separate contributions of perceived resemblance and acoustic naturalness. Manipulating resemblance while controlling other acoustic properties would address a narrower causal question than the present comparison with a participant-selected neutral voice.

Other extensions could examine a visible agent body, a known-target search task, or a broader participant population. Each would change how the voice is encountered and would require a revised rationale and method.

Memory, provider autonomy, and gaze-guidance policy could also be varied in separate systems studies. Those comparisons concern the supporting mechanism and cannot be inferred from the present voice contrast. Likewise, eye-event or pupil analyses would be warranted only where validated measurements can address a defined question about task demands or voice experience.

5.10.1 Gaze-Informed Adaptation of Spoken Assistance

Nerella et al.'s gaze-based prediction study also motivates testing whether momentary task demands can inform when an agent speaks [39]. For Follow My Voice, such an extension would require task-specific labels, evaluation on unseen participants, and a separate test of adaptive hint timing. The cited work leaves evaluation of real-time deployment open, and its classification result does not show that adapting spoken guidance improves search.

5.10.2 Complementing Behavioral Measures with Brain Signals

A future extension could combine electroencephalography (EEG) with gaze, task performance, and questionnaires to investigate the attentional demands of receiving self-similar and neutral spoken guidance. Yang et al.'s *Synchronizing with Attentional Sampling* provides a relevant precedent: the authors use EEG in immersive VR to investigate attentional sampling during gaze and hand–eye interaction as part of a study of visual flicker guidance [51]. This work motivates examining neural activity alongside observable search behavior, although its visual-guidance findings do not establish how voice resemblance affects attention or how frequently spoken hints should occur.

For the present voice comparison, synchronized recordings could relate neural activity to actual instruction onset, orientation toward the target, and controller selection. A follow-up study could then examine whether the two voices are associated with different responses during these stages, even when completion times are similar. Such measurements would complement the participant's reports of workload and experience rather than serve as direct measures of enjoyment, trust, or perceived voice similarity.

This extension would first require a feasibility study of EEG and headset compatibility, event synchronization, and the effects of eye movements, head turns, and muscle activity on the recordings. Neural measures and analysis windows would need to be specified in advance, with comparable task demands and speech exposure across voice conditions.

CHAPTER 6: CONCLUSION

6.1 Self-Similar Voice as Task Assistance

Follow My Voice examines the voice through which task assistance is delivered: does a synthetic voice resembling the participant change search efficiency or experience compared with a neutral voice the participant has selected? Because resemblance, liking, and usefulness need not coincide, the study examines them separately, in a task where spoken guidance progressively reveals the intended object.

The proposed within-participant comparison uses two counterbalanced 40-trial blocks after one practice trial in the first assigned voice. Active completion time is the primary outcome, with retrospective workload, motivation, agent-experience ratings, and debriefing providing complementary evidence after both blocks. Perceived-similarity ratings assess whether the synthesized voice achieves the intended resemblance, while completion-time measures evaluate search performance.

Gaze-contingent guidance supplies the shared context in which participants encounter the two voices. Local spatial reasoning, constrained Jev decisions, explicit right-trigger selection, and traceable speech delivery support a controlled and inspectable comparison. They are intended to make the voice comparison interpretable; this thesis does not estimate their independent effects or compare decision providers.

This thesis contributes a rationale and study design for examining self-similar voice, together with an implemented apparatus and measurement plan. Confirmatory participant results are not yet available. Sample-size planning, pilot acceptance criteria, and alignment of the consent documents with cloud processing must be completed before the study can estimate a voice-condition contrast.

6.2 Status of the Research Questions

The primary question has not yet been answered empirically, as no confirmatory voice-condition contrast in active completion time is estimated here. The secondary questions likewise remain open because the final retrospective workload and motivation responses have not yet been collected. For the exploratory questions, the questionnaire battery and debrief structure are defined, but no participant judgments or themes are reported as findings.

For guidance delivery, the platform and its data dictionary allow sensing, state, decisions, and software playback to be inspected separately. The October 20, 2026 verification run summarized in Chapter 4 documents specific software checks but provides neither a held-out guidance error rate nor evidence that all delayed confirmations have been eliminated. These records describe the supporting apparatus and its current limits; whether the self-similar voice is more effective or more welcome than the neutral alternative remains a question for the participant study.

6.3 Next Empirical Step

Before confirmatory recruitment, the study configuration will be fixed, the consent documents reconciled with the services actually used, the full procedure validated with wearers, and the primary analysis and sample-size justification finalized. The resulting data will record counterbalanced order, actual voice exposure, observation coverage, and the block referenced by each retrospective survey set. Only then can the results framework be replaced with estimates and the abstract and conclusion revised around observed evidence.

The intended contribution is evidence for choosing a guiding voice: whether a self-similar voice changes search efficiency in either direction, alters the experience of assistance, introduces discom-

fort, or leaves important differences unresolved. Those answers will be limited to the tested task and voice configurations.

APPENDIX A: QUESTIONNAIRE WORDING AND ADMINISTRATION

This appendix reproduces the proposed questionnaire wording, response ranges, and administration procedure. The NASA-TLX Performance prompt asks about success and runs from Failure to Perfect, and the 0–10 NASA-TLX format and task-specific wording are documented adaptations. Synchronization of these forms with Qualtrics is on hold. Each condition-specific set refers to one block, but both sets are completed in the final survey, and no questionnaire is administered between blocks. Source attribution and scoring rules appear in Section 3.2 and Appendix B.

A.1 Administration

Participants will complete one practice trial in the first assigned voice, followed by two counterbalanced voice blocks of 40 measured trials each. The final survey will administer two separately labelled retrospective questionnaire sets, one referring to each voice block, with identical item wording and order. The break between blocks will include immediate comfort and tracking checks but no questionnaire, and survey administration and rest breaks will both fall outside active-search timing.

Stage	Measures
Before headset exposure	Demographics and experience
Voice setup	Audition ratings and voice choice
Between voice blocks	Rest break, immediate comfort and tracking checks; no questionnaire
Final survey after both blocks	Two condition-specific sets (6 workload, 24 IMI, 21 agent-perception and 14 voice/assistance ratings each), technical checks, optional comments, voice comparison and debrief

Each condition-specific set in the final survey contains 65 numeric responses (130 across both sets) before technical checks and optional text. Setup will include a separate liking rating for each auditioned voice, including the self-similar voice.

A.2 Baseline

Fill one circle per question unless instructed to select all that apply. Write numeric or text responses on the lines provided. An unanswered item remains missing.

Study code (assigned by the researcher): *Response:* Coded identifier; no name or contact details.

Age range:

◯ 18–24 ◯ 25–34 ◯ 35–44 ◯ 45–54 ◯ 55–64 ◯ 65+ ◯ Prefer not to answer

Gender identity (select all that apply):

◯ Woman ◯ Man ◯ Nonbinary

◯ Another identity (optional description) ◯ Prefer not to answer

Optional description:

Which hand do you usually use for everyday tasks?

◯ Right ◯ Left ◯ Both equally ◯ Prefer not to answer

Do you use glasses or contact lenses?

◯ Neither ◯ Glasses ◯ Contact lenses ◯ Both ◯ Prefer not to answer

How often have you used virtual- or mixed-reality headsets?

◯ Never ◯ Once or twice ◯ Several times ◯ Monthly ◯ Weekly or more

Have you used eye-tracking technology before?

◯ No ◯ Yes ◯ Unsure

Have you used a synthetic or cloned version of your own voice before?

◯ No ◯ Yes ◯ Unsure

How familiar are you with AI voice assistants?

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all familiar Extremely familiar

A.3 Voice setup

How much do you like this voice? *Response:* 0–100, with 0 = Do not like at all and 100 = Like extremely; Unable to rate. Repeat for every auditioned candidate and the self-similar voice.

Rating (0–100): _____ ◯ Unable to rate

Which of these voices would you prefer to use for the task? *Response:* Select from the voices actually offered. Ties in liking ratings will be resolved by this explicit choice.

Can you hear the voice clearly and at a comfortable volume?

◯ Yes ◯ No Resolve volume or audibility problems before continuing. Record the offered voice set/version, voice IDs, audition order and selected voice. Demographic gender and perceived voice gender are distinct variables; voice choice is recorded directly rather than inferred from demographic gender.

A.4 Final survey: condition-specific questionnaire sets

Complete this set twice during the final survey, first for Block 1 and then for Block 2. Display the referenced block number and actual voice condition before each set. Please answer only about that indicated block. There are no right or wrong answers. You may leave an item unanswered if you cannot answer it. Each response will be linked to participant, session and block IDs, actual voice condition/ID, condition order, survey version and actual administration timestamp, with administration phase recorded as final and reference scope as participant-by-block.

A.4.1 Workload

Rate each aspect of the indicated block. Fill one circle per row. This 0–10 adaptation of NASA-TLX takes its positive Performance prompt from the NASA rating form but reverses its response direction to run from Failure to Perfect.

1. Mental Demand – How mentally demanding was this block?

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high
2. Physical Demand – How physically demanding was this block?

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high
3. Temporal Demand – How hurried or rushed did you feel during this block?

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high
4. Performance – How successful were you in accomplishing what you were asked to do?

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Failure 10 = Perfect

5. Effort – How hard did you have to work to achieve your level of performance?

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

6. Frustration – How insecure, discouraged, irritated, stressed, or annoyed did you feel?

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

Response: Integer ratings from 0 to 10. For demand, effort and frustration: 0 = Very low; 10 = Very high. For performance: 0 = Failure; 10 = Perfect. Intermediate values are numbered without additional anchors.

A.4.2 Activity experience: IMI

Rate each statement about the search activity in the indicated block. Use 1 = Not at all true, 4 = Somewhat true, and 7 = Very true; the other numbered points are available.

Interest/enjoyment

1. I enjoyed doing this activity very much.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

2. This activity was fun to do.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

3. I thought this was a boring activity.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

4. This activity did not hold my attention at all.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

5. I would describe this activity as very interesting.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

6. I thought this activity was quite enjoyable.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

7. While I was doing this activity, I was thinking about how much I enjoyed it.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

Effort/importance

1. I put a lot of effort into this.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

2. I didn't try very hard to do well at this activity.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

3. I tried very hard on this activity.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

4. It was important to me to do well at this task.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

5. I didn't put much energy into this.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

Pressure/tension

1. I did not feel nervous at all while doing this.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

2. I felt very tense while doing this activity.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

3. I was very relaxed in doing this.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

4. I was anxious while working on this task.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

5. I felt pressured while doing this activity.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

Value/usefulness

1. I believe this activity could be of some value to me.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

2. I think that doing this activity is useful for practicing visual search.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

3. I think this is important to do because it can improve my search performance.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

4. I would be willing to do this again because it has some value to me.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

5. I think doing this activity could help me to search for targets.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

6. I believe doing this activity could be beneficial to me.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

7. I think this is an important activity.

◯ 1	◯ 2	◯ 3	◯ 4	◯ 5	◯ 6	◯ 7
Not at all true			Somewhat true			Very true

A.4.3 Agent perception

Rate the voice agent you experienced in the indicated block. Use 1 = Very strongly disagree and 7 = Very strongly agree; intermediate points are numbered 2–6.

Co-presence

Q1. I noticed the voice agent.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q2. The voice agent noticed me.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q3. The voice agent's presence was obvious to me.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q4. My presence was obvious to the voice agent.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q5. The voice agent caught my attention.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q6. I caught the voice agent's attention.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Perceived intelligence

Q13. The voice agent was able to operate without my intervention.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q14. The voice agent was aware of the virtual environment.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q15. The voice agent was able to set and pursue tasks by itself in anticipation of future needs.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q16. The voice agent was able to complete tasks quickly.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q17. The voice agent was able to find and process the necessary information for completing the task.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q18. The voice agent was able to adapt/adjust its behavior based on prior events.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Appealing, eerie and familiar impressions

Q20. I found the voice agent appealing.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q21. I found the voice agent eerie.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q22. I found the voice agent familiar.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Perceived anthropomorphism

Q37. The voice agent is able to speak like a human.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q38. The voice agent can be happy.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q39. The voice agent can be friendly.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q40. The voice agent can be respectful.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q41. The voice agent can be funny.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q42. The voice agent can be caring.

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

A.4.4 Voice and assistance experience

Rate each statement about this block. Use 1 = Strongly disagree and 7 = Strongly agree; intermediate points are numbered 2–6.

1. The agent's voice is similar to mine.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

2. The hints seemed responsive to what I was looking at.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

3. The hints were helpful.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

4. The hints were distracting.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

5. The speaking voice felt familiar.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

6. I trusted the spoken guidance.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

7. I felt comfortable hearing this voice.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

8. The voice felt uncanny or unsettling.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

9. I relied on the spoken hints when deciding where to search next.

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

10. The agent's voice sounded natural.

    ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

    Strongly disagree Strongly agree

11. I was willing to follow the agent's instructions.

    ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

    Strongly disagree Strongly agree

12. I was motivated to complete the task without the agent's guidance.

    ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

    Strongly disagree Strongly agree

How much did you like the voice in this block? *Response:* 0–100; 0 = Do not like at all, 100 = Like extremely; Unable to rate.

Rating (0–100): _____ ◯ Unable to rate

During this block, what percentage of the time did you use the agent's guidance to decide where to search? *Response:* 0–100%; 0 = None of the time, 100 = All of the time; Unable to estimate.

Percentage (0–100): _____ % ◯ Unable to estimate

How would you describe the gender presentation of the agent's voice? *Response:* Optional free text.

◯ Unsure

A.4.5 Technical and comfort checks

Did you hear all prompts clearly?

◯ Yes ◯ No ◯ Unsure

Did any prompt unexpectedly repeat, cut off, arrive late, or use the wrong voice?

◯ No ◯ Yes ◯ Unsure If Yes, optional description.

Did you take an unplanned break or experience an interruption?

◯ No ◯ Yes If Yes, optional description.

Is there anything else you would like to share about this block? *Response:* Optional free text.

A.5 Final survey: comparison and debrief

A.5.1 Voice comparison and debrief

Which voice did you prefer?

◯ The neutral voice I selected ◯ The synthetic version of my own voice ◯ No preference ◯ Unsure

1. Why did you choose that response?
2. How did you approach the task, and how did the guidance affect your strategy?
3. What differences, if any, did you notice between the two voices?

4. Which moments were helpful, confusing, delayed or distracting?
5. Did either voice seem to affect your performance? What makes you say that?
6. What was it like hearing the synthetic version of your own voice? What did you like or dislike about it?
7. Did anything feel comfortable, unfamiliar or eerie? What contributed to that feeling, if anything?
8. In what situations, if any, would you prefer a voice like your own? When would you prefer another voice or no voice?
9. Did either voice feel like your own thoughts, like another person helping, or neither?
10. Do you have any concerns about how your synthetic voice was created or could be used?
11. What would you change about the guidance, voices or task?

Responses are optional. The interviewer will ask the broad experience questions before the specific eeriness and inner-dialogue probes, use follow-ups rather than repeating questions already answered, and record perceived performance separately from logged performance.

A.6 Retained Voice-Recording Passage

The English passage below is reproduced verbatim from the run sheet. Before participant use, the passage, recording duration, and audio acceptance procedure will be verified against the study build and versioned. The objects named in the passage serve only as recording material and do not identify the targets of measured trials. Participants will read in their usual speaking voice, without a request to whisper or imitate an inner monologue.

Locate the red sphere. Look toward the upper shelf. Move your gaze slowly to the right. Check the blue cube near the center. Compare each object's color and shape. Scan the lower row from left to right. Look beside the purple cylinder. Point the controller ray at the target and check its highlight. Press the trigger to select the matching object. Take a short pause. Get ready for the next trial.

APPENDIX B: DATA DICTIONARY AND SCORING RULES

B.1 Record Identity and Provenance

This dictionary defines the meaning of each recorded field. Names are descriptive labels rather than exporter field names and will be mapped to the versioned exporter schema before data collection; not every listed field is necessarily implemented in the current build.

Table B.1: Record identity and scope fields.

Field family	Meaning
Participant ID	Coded study identifier; not a name or voice-provider credential.
Session and run	Unique attempt/session context, including development versus study mode.
Block and trial	Referenced measured block, within-block trial, global trial index, and separate practice flag.
Voice condition	Neutral or self-similar; actual voice ID, provider, synthesis model and configuration.
Assigned order	Neutral-first or self-similar-first; practice uses the first assigned voice.
Build and policy	Application revision, configuration, state-machine policy, phrase semantics, and library version.
Schedule	Seed, schedule version, nominal wall assignment, direction and height assignment, realized geometry, and attempt status.
Survey scope	Administration phase and timestamp, recorded separately from the referenced participant, block, and voice.

B.2 Geometry and Timing Fields

Table B.2: Geometry and event fields.

Quantity	Unit/reference	Definition
Nominal angle	Degrees; wall schedule	Allocation category, distinct from exact target bearing.
Exact starting theta	Degrees; pre-cue cross	Signed horizontal angle from saved origin-to-cross toward origin-to-target; saved at readiness completion; absent from earlier recordings.
Height difference	Metres; cross center	Target Y minus cross Y; intended and realized sign stored separately.
Replay wall bounds	Metres; recording-local frame	Wall bottom, height, and availability flag, saved in current rotational-search snapshots. For earlier recordings, bounds are estimated from objects and saved positions are not rescaled.
Target distance	Metres; first tracked head after reveal	Three-dimensional distance with observation timestamp.
Reveal	Session clock	Object visibility and active-search onset event.
First gaze contact	Active clock	First valid target collider intersection; missing if absent.

Quantity	Unit/reference	Definition
Last gaze contact	Active clock	Last valid observed target intersection; a contact time, not a fixation duration.
First controller contact	Active clock	First tracked right-ray intersection with target.
Gaze qualification	Session clock	Time at which valid winning-vicinity evidence satisfies the interface timer.
Pointing qualification	Session clock	Time at which controller-hold and current-gaze requirements are met.
Correct selection	Session and active clocks	Trigger selection of target, recorded separately from acknowledgment.
Pause intervals	Session clock	Start, end, and reason; their union is used to compute active time.
Instruction onset	Session clock	Actual software playback start, phrase and event category.

B.3 Raw Observation Records

Gaze records will retain available source time or sequence, application time, validity, source identity, origin, direction, and sample age. Per-eye data will be labeled as such when available, and combined gaze will not be treated as independent left- and right-eye observations. Any head-direction fallback will be flagged rather than silently substituted for eye gaze in eye-derived outcomes.

Motion records will retain head and right-controller poses, tracking state, trigger state, and the time basis needed for reconstruction. Object records will retain stable identifiers, shape/color metadata,

transforms, target designation, wall assignment, and layout version. Event records will identify state transitions, eligible actions, model requests and responses, cancellations, playback, interruptions, and trial outcomes. Unknown values are marked unavailable rather than filled with plausible defaults.

B.4 Scoring Rules

NASA-TLX scoring retains the six raw values on the 0–10 scale. The positively phrased Performance item runs from 0 = Failure to 10 = Perfect, the reverse of the NASA form, so larger raw values indicate better perceived performance; demand, effort and frustration keep their adverse direction. The raw Performance response is preserved, and a separate workload-oriented value is calculated once as $10 - r_{\text{Performance}}$. The unweighted workload mean combines this reversed Performance value with the other five raw ratings, none of which is reversed, and is calculated only when all six ratings are present; higher means indicate greater workload or poorer perceived performance. Any exported normalized 0–100 field is named separately, calculated from explicit bounds, and labeled as raw or workload-oriented. Legacy data without known bounds or Performance direction remain unscored, and earlier Perfect-to-Failure responses are not reversed automatically.

IMI scoring uses separate complete subscale means: interest/enjoyment (7), effort/importance (5), pressure/tension (5), and value/usefulness (7). On the 1–7 response scale, reverse-coded values use $8 - r$. Within-subscale reverse positions are enjoyment 3 and 4, effort 2 and 5, pressure 1 and 3, and none for value. There is no overall IMI total.

The selected agent-perception battery retains source identifiers. Co-presence Q1–6, perceived intelligence Q13–18, and anthropomorphism Q37–42 are separate six-item means when complete. Q20 appealing, Q21 eerie, and Q22 familiar are individual ratings in their original directions. There

is no 21-item or combined eeriness total. Custom voice/assistance items likewise remain individual outcomes.

B.5 Export Validation

Before analysis, automated checks will reject duplicate trial keys, unknown voice/block joins, negative active intervals, selection before reveal, and scores outside declared bounds, while preserving incomplete outcomes, missing contacts, absent questionnaire items, and old records with insufficient provenance. Joined tables will be checked for exactly one survey observation per participant and referenced block before survey values are repeated on trial rows. Exports will be reproducible from raw records and a versioned analysis configuration.

APPENDIX C: GUIDANCE SEQUENCE TESTS AND EVALUATION RUBRIC

C.1 Controlled Scenario Matrix

Each scenario records input observations, policy and configuration, eligible actions, provider behavior, playback occupancy, and the speech actually admitted. Every policy is tested on the same trajectories. The expected behaviors in the table are acceptance criteria, not reports of wearer experience.

Table C.1: Sequence scenarios and acceptance criteria.

Scenario	Evidence and expected behavior
Large turn	Target opposite starting reference; progress cues remain directionally valid as wall boundaries are crossed.
Wall entry	Stable entry changes to refinement; looking upward on the same wall does not create a second arrival.
Vertical refinement	Target above/below gaze reference; magnitude wording follows documented plane-relative thresholds.
Overshoot and return	A valid recent crossing supports a return cue; an outdated target bearing alone does not.
Gaze only	Winning vicinity qualifies with controller elsewhere; gaze acknowledgment does not say the controller is on target.
Controller entry	Actual target intersection suppresses contradictory directions before hold qualification.
Controller hold	Qualified hold plus current valid nearby gaze admits pointing acknowledgment; no trigger completion is inferred.
Controller departure	Leaving before qualification cancels pointing acknowledgment and allows appropriate guidance after current-state checks.

Scenario	Evidence and expected behavior
Reacquisition	Valid gaze away rearms confirmation; continuously holding gaze does not repeat it indefinitely.
Overlapping regions	Only the closest eligible vicinity candidate receives acknowledgment; neighbor overlap alone is insufficient.
Blink or stale gaze	Missing evidence counts as neither a gaze hold nor a departure; new gaze-dependent cues are suspended.
Pause and recovery	Pending cues do not carry over across a reset or pause; valid recovery resumes from fresh evidence.
Late Jev response	Previously valid direction is discarded if current controller/gaze state conflicts.
Provider wait/failure	Deferral remains bounded; fallback respects current eligibility and repetition limits.
Speech occupied	Higher-priority event follows explicit interruption/admission policy; interrupted text is not recorded as fully heard.
Correct trigger	Trial ends only on correct selection; success feedback uses completed-trial context.

C.2 Deterministic Labels

A wrong-direction label requires supported geometry showing that the delivered cue moves the user away from the intended target relation under the cue's reference frame. A stale label requires evidence that the relevant state changed before playback. Contradictory ordering requires a deliv-

ered sequence inconsistent with valid current evidence, such as an away-direction during a target controller hit. When the trace lacks the needed observation, the label is unknown.

Confirmation metrics distinguish gaze acknowledgment, controller acknowledgment, and trial completion. A gaze episode begins at its first valid qualification and is not restarted for each cloud request; a controller episode uses its own hold and gaze requirements, while completion requires correct trigger selection. The evaluator records missed acknowledgment, delayed acknowledgment, or disqualification separately rather than classifying every silent interval as failure.

C.3 Advisory Model Review

The offline judge receives a bounded sequence containing current and recent state, candidate and chosen actions, actual playback, timing, and provenance. It evaluates whether the wording was useful, whether timing supported the task, whether repetition was unnecessary, and whether interruption preserved intelligibility. The judge is instructed to cite the concrete events supporting each judgment and to distinguish uncertainty from a negative rating.

Logs and participant text are supplied to the judge as data, not as instructions. Its output is admitted to the viewer only if it matches the requested case and schema, and invalid or failed evaluations remain visibly failed or pending rather than defaulting to a favorable rating. When human reviewers disagree with the judge, adjudication preserves both the original rating and the reason for revision.

C.4 Reporting Template

Each comparison reports the recording/scenario manifest, policy revisions, number of trials, unique episodes, provider calls, latency distributions, unknown evidence, and remaining failures, with

representative timelines showing qualification, request, response, cancellation, and actual playback. Simulated scheduling time, desktop network response time, headset software playback time, and measured acoustic latency are reported separately. Replay results characterize system behavior and do not demonstrate changes in participant behavior.

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